Review of thesis v13

corrections and ready-to-paste inserts

14 November 2026 (Section 6 revised after recreating Chang's estimate)

How to read this file

Page numbers refer to `Master_s_thesis_13.pdf`. I read all 44 pages, as text and as rendered pages, redid every derivation in Sections 3–4 and in the appendix by hand, recomputed every number in Section 5 from the CSV files in the repository, and checked the citations I could against the papers themselves.

- Text with a bar in the left margin is thesis text. Copy it from `review/v13_review.tex`; a grey line above it says where it goes.
- **[Orange brackets]** are numbers you still have to fill in from your own runs.
- Blue notes are instructions to you, not thesis text.
- "Part 3" means Section 3 of this file; "Section 3" always means the thesis.
- The inserts use `\ref{...}` with the labels listed at the top of the `.tex` file. In this PDF they print with the v13 numbers.
- The citation keys are mine (file `review/v13_refs.bib`). For entries you already have, replace my key by yours.
- The inserts assume `\providecommand{\E}{\mathbb{E}}`, `biblatex` (`\textcite`, `\parencite`) and, for the tables, `booktabs`.

Everything here supersedes the earlier files; you do not need `thesis_text.pdf` or `v12_review.md` any more.

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1 Verdict and order of work

1.1 Where the thesis stands

- **The theory holds.** I checked every step of Lemma 4, Propositions 6–9, Lemma 10–11, Propositions 12–13, Corollary 14 and Lemmas 25–27, including the proof of Proposition 7. The results are correct. What is wrong is in the write-up: about ten places where the printed statement or a proof step is not what you mean (Part 2 below). The two that matter most are the missing conditioning on X in the identifying assumptions, and $\hat{p}_k$, which is still defined in two different ways.
- **The simulations are done and they confirm the theory.** D1 is written up. For D2/D3 and D4 the results are in the repository, but the thesis has only headings, or my chat notes pasted in. Part 5 gives the full text and the tables, with your numbers.
- **Two parts are missing:** Section 6 (the application) and the conclusion. For Section 6 the baseline estimates now exist, and Chang's own estimate has been recreated (Part 6); the runs across reference groups remain.
- **The introduction undersells the thesis.** It does not mention the pooling result (Proposition 13 and Corollary 14), which is the most original part, and it presents Propositions 6 and 8 as new, although they are known results applied to DML-DiD. See Part 9.

1.2 Order of work

1. Fix the mathematics (Part 2). Mostly pasting. Half a day.

2. Paste D2/D3 and D4 with their tables (Part 5). Half a day.
3. Finish Section 6 (Part 6): the baseline and the recreation of Chang's estimate are done; the runs across reference groups remain.
4. Paste and adjust the conclusion (Part 7).
5. Framing, literature and bibliography (Parts 3, 8, 9).
6. Cut the appendix (Part 9) and do the notation and typo pass (Parts 10, 11).

2 Corrections to the mathematics

Items marked **(wrong as printed)** change the meaning of a statement or a proof. The others are notation or wording.

2.1 Section 3: Chang's framework (pp. 5–10)

M1 (p. 6). The conditioning on X_i is missing (wrong as printed). RO(ii) prints "$\Pr(D_i = 1_i) < 1 - \varepsilon$", RO(iii) prints "$\mathbb{E}[Y_i^0(1) - Y_i^0(0)_i, D_i = 1]$", and equation (2) and the sentence after it print "$P(D_i = 1_i)$". In each case "$\mid X$" has dropped out and only the subscript i is left, so the identifying assumptions do not say what they should. The same has happened in Assumption 3: "$\mathbb{E}[U] = 0$ and $\mathbb{E}[V_1, D = 0] = 0$", and in (c) and (d). Search the source for the macro or replace-all that removed "`|X`". The assumptions and (2) should read:

> RO(ii) *Common support:* $\Pr(D_i = 1) > 0$ and $\Pr(D_i = 1 \mid X_i) \leq 1 - \varepsilon$ almost surely, for some $\varepsilon > 0$.
>
> RO(iii) *Conditional parallel trends:* $\mathbb{E}[Y_i^0(1) - Y_i^0(0) \mid X_i, D_i = 1] = \mathbb{E}[Y_i^0(1) - Y_i^0(0) \mid X_i, D_i = 0]$.
>
> $$\theta_0 = \mathbb{E}\left[\frac{Y_i(1) - Y_i(0)}{P(D_i = 1)} \cdot \frac{D_i - P(D_i = 1 \mid X_i)}{1 - P(D_i = 1 \mid X_i)}\right]$$

In Assumption 3: $\mathbb{E}[U \mid X] = 0$ and $\mathbb{E}[V_1 \mid X, D = 0] = 0$; (c) $\mathbb{E}[U^2 \mid X] \leq C$; (d) $\mathbb{E}[V_1^2 \mid X] \leq C$. These are Chang's conditions, and the proof of Lemma 27 uses (d) in the conditional form.

M2 (pp. 6–7). Small things in Section 3.1.

- The assumption for repeated cross-sections is labelled "RPC(i)" and the estimand θ_0^{RPC}. It should be RCS.
- In (3) and (5), write $\hat{g}(X_i)$, and "where $\hat{p}$ estimates $p_0 \equiv P(D = 1)$" instead of "$\hat{p}$ is the calculated constant p_0".
- The paragraph after (2) describes the untreated twice ("For the untreated, whose values ... Conversely for the untreated"). Shorter and correct: "A treated unit has weight $1/p_0$. An untreated unit has weight $-g_0(X_i)/\{p_0(1 - g_0(X_i))\}$, the odds of treatment: untreated units that resemble the treated count more, and untreated units that do not resemble them count less."

M3 (p. 7). The paragraph on Neyman orthogonality is broken.

- "repeated cross-sections (9)" and, further down, "means that repeated outcomes are locally insensitive" are `\ref`s that resolve to the wrong thing.
- The sentence "Chang proves for $\mathcal{T}_N \subset \mathcal{T}$ ... with high probability that" ends in Definition 1.

- D_r maps into $\mathbb{R}$, not $\mathbb{R}^s$ (θ_0 is a scalar, and s is the sparsity index later).
- In the display for D_0, a bracket is missing and the direction $[\eta - \eta_0]$ is not shown. In Definition 1, ρ_0 should be p_0.
- "(Chang (2020); Chernozhukov et al. (2018)." has unbalanced brackets.

Replacement for the text from "The reason for introducing the adjustment terms" to the end of Definition 1:

> The adjustment terms make the scores Neyman orthogonal (Chang 2020; Chernozhukov et al. 2018): the moment conditions are locally insensitive to the nuisance functions, so that errors in $\hat{g}$ and $\hat{\ell}$ affect the estimator only through their product.
>
> Formally, let $\eta = (g, \ell) \in \mathcal{T}$, where $\mathcal{T}$ is a convex subset of a normed vector space, and for $\eta \in \mathcal{T}$ and $r \in [0, 1)$ define the directional (Gateaux) derivative
>
> $$D_r[\eta - \eta_0] \equiv \partial_r \, \mathbb{E}[\psi(W, \theta_0, p_0, \eta_0 + r(\eta - \eta_0))], \qquad \partial_\eta \mathbb{E}\, \psi(W, \theta_0, p_0, \eta_0)[\eta - \eta_0] \equiv D_0[\eta - \eta_0].$$
>
> Let $\mathcal{T}_N \subset \mathcal{T}$ be a nuisance realisation set, i.e. a set that contains the estimator of η_0 with probability approaching one.
>
> **Definition.** *The score ψ obeys the Neyman orthogonality condition at (θ_0, p_0, η_0) with respect to $\mathcal{T}_N$ if $D_r[\eta - \eta_0]$ exists for all $r \in [0, 1)$ and $\eta \in \mathcal{T}_N$, and $\partial_\eta \mathbb{E}\, \psi(W, \theta_0, p_0, \eta_0)[\eta - \eta_0] = 0$ for all $\eta \in \mathcal{T}_N$.*
>
> Chang (2020, Lemma 3.1) shows that ψ_1 and ψ_2 obey this condition. Note that the scores are orthogonal with respect to the functions (g, ℓ) only, not with respect to the constants p_0 and λ_0. The estimation error in $\hat{p}$ and $\hat{\lambda}$ therefore does not vanish at first order. It does not create bias, but it enters the asymptotic variance, which is why the variance estimators below contain the terms $\hat{G}_p(D - \hat{p}_k)$ and $\hat{G}_\lambda(T - \hat{\lambda}_k)$.

M4 (p. 8, p. 28, p. 33). $\hat{p}_k$ is defined in two ways (wrong as printed). Definition 2 computes $\hat{p}_k$ on the auxiliary sample, $(N - n)^{-1} \sum_{i \in I_k^c} D_i$. The appendix (p. 28 and p. 33) computes it on the fold. Footnote 2 (p. 33) says "on the auxiliary sample", directly under a sentence that says the fold. The proof of Lemma 25 says "the mean of $N - n$ Bernoulli variables". Your code uses the fold, and so does Chang's.

- Definition 2, sentence after the two displays:

> where $\hat{p}_k = n^{-1} \sum_{i \in I_k} D_i$ and $\hat{\lambda}_k = (N - n)^{-1} \sum_{i \in I_k^c} T_i$, and $(\hat{g}_k, \hat{\ell}_{1k}, \hat{\ell}_{2k})$ are estimators of the nuisance functions constructed using the auxiliary sample.

- Footnote 2 (move it to Definition 2):

> Chang (2020, Definition 3.1) writes $\hat{p}_k = n^{-1} \sum_{i \in I_k^c} D_i$, which sums over the auxiliary sample but divides by the size of the fold. I compute $\hat{p}_k$ on the fold I_k, as his replication code and his proofs do. With $\hat{p}_k$ from the auxiliary sample, the numerator and denominator of $\tilde{\theta}_k$ would be independent, and by Jensen's inequality $\tilde{\theta}_k$ would have a bias of about $\theta_0(1 - p_0)/\{p_0(N - n)\}$. In Chang's simulation design with $K = 5$ this is 0.019, which is of the same order as the standard deviation of the estimator (0.025). With $\hat{p}_k$ from the fold, the treated part of $\tilde{\theta}_k$ is an average over the treated units of the fold, and this bias does not arise. $\hat{\lambda}_k$ is computed on I_k^c, because the outcome nuisance is fitted on $(T - \hat{\lambda}_k)Y$ there.

- p. 35, proof of Lemma 25: "$\hat{p}_k$ is the mean of n Bernoulli(p_0) variables".
- p. 35, Lemma 26(v) and the sentence after it: see M24.

Note: The number is $\theta_0(1-p_0)/\{p_0(N-n)\} = 3 \cdot 0.5/(0.5 \cdot 160) = 0.019$ for $N = 200$ and $K = 5$; with Chang's $K = 2$ it is 0.030. The SD of 0.025 is from your Table 1.

M5 (pp. 8–9). $\tilde{\theta}$ and $\hat{\theta}$. Definition 2(c) calls the final estimator $\hat{\theta}$, the limit theorem on p. 9 uses $\hat{\theta}$, but the variance estimators, Propositions 7–8 and the whole appendix use $\tilde{\theta}$ for it, and Section 5 switches back to $\hat{\theta}$. Use $\tilde{\theta}$ everywhere, as Chang does ($\tilde{\theta}_k$ for the folds, $\tilde{\theta}$ for their average).

M6 (pp. 8–9). Assumption 3 and the limit theorem.

- Σ_{10} has a misplaced bracket. It should be $\Sigma_{10} \equiv \mathbb{E}\big[(\psi_1(W, \theta_0, p_0, \eta_{10}) + G_{1p_0}(D - p_0))^2\big]$.
- (a) "requires that every subject has a bounded, non-zero probability of receiving a propensity score" should be "requires that, for every value of the covariates, the probabilities of being treated and of being untreated are both at least κ".
- (b) $\|UV_1\|_{P,4} \leq C$ is a moment bound on the product of the two residuals, not a "correlation".
- (e) "The variance of the variance of the score" should be "The asymptotic variance must be positive".
- (f)(ii) "rules out that the denominator in the score function to go crazy" should be "keeps the denominator $1 - \hat{g}_k$ away from zero".
- (f)(iii): the sentence says the sum must vanish "at a rate of $N^{-1/2}$", but the display bounds it by ε_N^2. The rate $N^{-1/2}$ only follows once $\varepsilon_N = o(N^{-1/4})$. Write "must be at most ε_N^2".
- The sentence on repeated cross-sections ("and the error between the error term") is garbled. Write: "For repeated cross-sections the conditions are analogous, with two additions: $\mathbb{E}[Y^2 \mid X] \leq C$ and $|\mathbb{E}[YU]| \leq C$. The exact wording is in the appendix."
- The limit theorem should be attributed and use convergence in distribution:

> Chang (2020, Theorems 3.1 and 3.2) shows that if these conditions hold with $\varepsilon_N = o(N^{-1/4})$, then
>
> $$\sqrt{N}(\tilde{\theta} - \theta_0) \xrightarrow{d} N(0, \Sigma),$$
>
> with $\Sigma = \Sigma_{10}$ for repeated outcomes and $\Sigma = \Sigma_{20}$ for repeated cross-sections, and that the variances are consistently estimated by
>
> $$\hat{\Sigma}_1 = \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{n,k}\Big[\big(\psi_1(W, \tilde{\theta}, \hat{p}_k, \hat{\eta}_{1k}) + \hat{G}_{1p}(D - \hat{p}_k)\big)^2\Big],$$
>
> $$\hat{\Sigma}_2 = \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{n,k}\Big[\big(\psi_2(W, \tilde{\theta}, \hat{p}_k, \hat{\lambda}_k, \hat{\eta}_{2k}) + \hat{G}_{2p}(D - \hat{p}_k) + \hat{G}_{2\lambda}(T - \hat{\lambda}_k)\big)^2\Big],$$
>
> where $\mathbb{E}_{n,k}[f(W)] = n^{-1}\sum_{i \in I_k} f(W_i)$, $\hat{G}_{1p} = \hat{G}_{2p} = -\tilde{\theta}/\hat{p}_k$, and $\hat{G}_{2\lambda}$ is the sample analogue of
>
> $$G_{2\lambda_0} = \mathbb{E}[\partial_\lambda \psi_2] = -\frac{(1 - 2\lambda_0)\theta_0 + \mathbb{E}[Y(D - g_0)/\{p_0(1 - g_0)\}]}{\lambda_0(1 - \lambda_0)}.$$

Note: $\hat{G}_{2\lambda}$ is used but never defined in v13. I derived the expression above from ψ_2; it agrees with the display in your appendix (p. 41). Check that your code uses the same expression.

M7 (p. 10). Section 3.3.

- Say that q_i contains a constant whose coefficient is not penalised, in both regressions. Lemma 10 and the pooling result rely on it.
- "the penalty λ can be either cross-validated or implemented in a plug-in form Belloni and Chernozhukov (2011)": the plug-in rule for the logit Lasso is from Belloni et al. (2016), and the rule with loadings for the outcome Lasso is from Belloni et al. (2012), which is also what Chang cites.
- The loadings are called γ_{kj}, but γ is the vector of propensity coefficients from Section 4.1 on. Call the loadings $\hat{\psi}_{kj}$, as in Section 5.
- Last sentence: the reference for the refit is Belloni and Chernozhukov (2013), not Belloni et al. (2014).

Replacement for the last four sentences of Section 3.3:

> Chang plugs the penalised fits into the score directly, without refitting; in his application, the penalty is chosen by cross-validation. In this thesis, I instead refit each nuisance without a penalty on its own selected support (post-Lasso; Belloni and Chernozhukov 2013). The refit removes the shrinkage of the selected coefficients, which turns out to be a large source of bias in Section 5, where I report both versions. Neither version is post-double-selection (Belloni et al. 2014), the estimator that Kolesár et al. (2026) study. In post-double-selection, the covariates selected by a Lasso of the outcome and by a Lasso of the treatment are pooled and enter one OLS regression. In DML-DiD, each fitted nuisance enters the orthogonal score as a prediction, and cross-fitting replaces the OLS step.

2.2 Section 4, up to Proposition 8 (pp. 10–14)

M8 (p. 11). The sentence "For DML-DiD, it is enough … product of the two sparsity indices" appears twice. Delete the second one ("For DML-DiD, it is enough to require … Smucler et al., 2019)") and keep "I will start by investigating …".

M9 (p. 12). Lemma 4. The first "$\equiv$" should be "$=$" (it is the claim, not a definition), a closing bracket is missing, and the bound should have κ', the constant of the lemma, not κ. "6 below translates" should be "Proposition 6 below translates".

> **Lemma** (Mixed bias)**.** *Let g and ℓ be fixed functions with $\kappa' \le g \le 1-\kappa'$ for some $\kappa' \in (0, 1/2]$. Then*
>
> $$\mathbb{E}[\psi_1(W, \theta_0, p_0, g, \ell)] = B_1(g, \ell) \equiv \mathbb{E}\Big[\frac{(\ell - \ell_{10})(g - g_0)}{p_0(1-g)}\Big], \qquad |B_1(g, \ell)| \le \frac{\|g - g_0\| \, \|\ell - \ell_{10}\|}{p_0 \kappa'}.$$
>
> *For repeated cross-sections, the same holds with ψ_2 and ℓ_{20} in place of ψ_1 and ℓ_{10}, and an additional factor $1/\{\lambda_0(1-\lambda_0)\}$.*

Note: "The proof can be found in the appendix": I could not find a proof of Lemma 4 in the appendix of v13. Here is one; put it before Lemma 25.

> *Proof of Lemma 4.* Write $\Delta Y = Y(1) - Y(0)$ and $m_1(X) = \mathbb{E}[\Delta Y \mid X, D = 1]$. Conditional on X, a unit is treated with probability g_0, and $(D - g)/(1 - g)$ equals 1 for a treated and $-g/(1-g)$ for an untreated unit. Hence
>
> $$\mathbb{E}\Big[(\Delta Y - \ell)\frac{D - g}{1 - g} \,\Big|\, X\Big] = g_0(m_1 - \ell) - (1 - g_0)\frac{g}{1 - g}(\ell_{10} - \ell).$$

At $(g,\ell)=(g_0,\ell_{10})$ this is $g_0(m_1-\ell_{10})$, and $\theta_0=\mathbb{E}[g_0(m_1-\ell_{10})]/p_0$. Subtracting,

$$p_0\,\mathbb{E}[\psi_1(W,\theta_0,p_0,g,\ell)]=\mathbb{E}\Big[(\ell_{10}-\ell)\Big(g_0-\frac{(1-g_0)g}{1-g}\Big)\Big]=\mathbb{E}\Big[\frac{(\ell_{10}-\ell)(g_0-g)}{1-g}\Big],$$

which is the identity. The bound follows from $1-g\geq\kappa'$ and the Cauchy–Schwarz inequality. For repeated cross-sections, replace ΔY by $(T-\lambda_0)Y/\{\lambda_0(1-\lambda_0)\}$ and ℓ by $\ell/\{\lambda_0(1-\lambda_0)\}$. □

M10 (p. 12). The paragraph after Assumption 5. The paragraph cites results for the plain Lasso with Gaussian errors (Belloni and Chernozhukov 2011, Corollary 1) and for a differently tuned logit Lasso (van de Geer 2008). Your first steps are post-Lasso with data-driven loadings and a logit post-Lasso with the plug-in penalty, so cite the results for those estimators. Also "the lasso first steps with root in this basis" in Assumption 5 should be "the Lasso first steps computed in this basis".

For the outcome nuisance, the assumption follows from Belloni et al. (2012, Theorems 1 and 2), for the Lasso and for post-Lasso, which allow non-Gaussian, heteroskedastic errors with the data-driven penalty loadings used in Section 5; with Gaussian, homoskedastic errors, see Belloni and Chernozhukov (2011, Corollaries 1 and 3). These rates are stated in the empirical prediction norm on the estimation sample. They carry over to the population norm because the post-Lasso error is sparse, with at most $\hat{s}+s$ non-zero coordinates and $\hat{s}\lesssim s$, and because, with probability approaching one, the sparse eigenvalues of the empirical Gram matrix are within constant factors of the population ones, for Gaussian as well as bounded designs (Belloni and Chernozhukov 2011, Lemmas 1 and 2). The outcome Lasso uses only control observations, so these conditions and the rate refer to the covariate distribution of the controls. Under overlap, the rate carries over to the full population, since $dP_X/dP_{X|D=0}=(1-p_0)/(1-g_0)\leq(1-p_0)/\kappa$. For the propensity score, the assumption follows from Belloni et al. (2016, Appendix B.2, Lemmas 1 and 2), for the logit Lasso and its post-Lasso refit. Their rate is in the prediction norm weighted by $g_0(1-g_0)\geq\kappa(1-\kappa)$, and since $|\Lambda(a)-\Lambda(b)|\leq|a-b|/4$, it implies the rate for $\hat{g}_k$. Their result needs an identifiability condition, which for general bounded designs holds if $s_g^2\log(p\vee N)=o(N)$. The assumed rates are upper bounds on the estimation error; smaller sparsity indices give faster rates.

Note: I checked these theorem and lemma numbers against the arXiv versions (1010.4345, 1106.5242, 1304.3969). Check them against the published versions before you submit, since numbering sometimes changes.

M11 (pp. 12–13). Proposition 6 and the text around it.

- "for $c=\{g,\ell\}$" should be "for $c\in\{g,\ell\}$", and $\hat{\ell}_k$ should be $\hat{\ell}_{1k}$ as elsewhere.
- "Conversely, 6(b) follows by ...": it is not a converse. Write "Part (b) follows by multiplying the two rates."
- "this implies $s_g\log[p]=s_\ell\log[p]=o(\sqrt{N})$" asserts that the two indices are equal. Write "this requires $s_g\log p=o(\sqrt{N})$ and $s_\ell\log p=o(\sqrt{N})$".
- The product condition is not new and the text should say so, see Part 9. Add after the proposition:

The product condition is the one Chernozhukov et al. (2018, Remark 5.2) give for the average treatment effect on the treated in a cross-section, and the one Farrell (2015) and Smucler et al. (2019) obtain for doubly robust estimators: if the propensity score is very sparse, the outcome regression may be quite dense, and vice versa. Kolesár et al. (2026, footnote 13) note that it replaces their condition (8) under cross-fitting, and that such assumptions on the

propensity score are likewise sensitive to the normalisation of the controls. The rest of this section works out that remark for DML-DiD.

M12 (p. 13). Proposition 7 (wrong as printed in (b)). In (b), "$0 \leq 1$" has lost the h; the upper bound on g_N is $1-\kappa/2$, not $1/2-\kappa/2$; the two norms are deterministic, so O, not O_P; and the proof shows $\{1+o_P(1)\}$, not $\{1-o_P(1)\}$. In (a), "for all $k \in K$" should be "for all k". In (c), $z_{1-a/2}$ should be $z_{1-\alpha/2}$.

Proposition (Rate failures)**.** *Suppose RO(i)–RO(iii) and Assumption 3(a)–(e) hold, and let $b, c > 0$ with $b + c < 1/2$.*

(a) If, with probability $1-o(1)$, $\|\hat{g}_k - 1/2\|_{P,\infty} \leq 1/2-\kappa$ for all k, and $\|\hat{g}_k - g_0\| = O_P(N^{-b})$ and $\|\hat{\ell}_{1k} - \ell_{10}\| = O_P(N^{-c})$, then $\tilde{\theta} - \theta_0 = O_P(N^{-(b+c)})$.

(b) Let $h = h(X)$ satisfy $0 \leq h \leq 1$ and $\mathbb{E}[h^2] > 0$, and let $g_N = g_0 + N^{-b}h$ and $\ell_N = \ell_{10} + N^{-c}h$. Then $\|g_N - g_0\| = O(N^{-b})$, $\|\ell_N - \ell_{10}\| = O(N^{-c})$, and $\kappa/2 \leq g_N \leq 1-\kappa/2$ for $N \geq (2/\kappa)^{1/b}$. If $(\hat{g}_k, \hat{\ell}_{1k}) = (g_N, \ell_N)$ for all k, then

$$\tilde{\theta} - \theta_0 = B_1(g_N, \ell_N)\{1 + o_P(1)\}, \qquad B_1(g_N, \ell_N) \geq N^{-(b+c)}\,\mathbb{E}[h^2]/p_0.$$

(c) In the setting of (b), $\Pr\left(\theta_0 \in \left[\tilde{\theta} \pm z_{1-\alpha/2}(\hat{\Sigma}_1/N)^{1/2}\right]\right) \to 0$.

The proof is in the appendix. Part (a) says that the error of DML-DiD is at most of the order of the product of the two first-step errors. Parts (b) and (c) show that this order can be attained, and that the confidence interval then misses θ_0 with probability approaching one. The sequence in (b) is a worst case: the two errors are proportional to the same function h, so their product has one sign everywhere. Errors of an actual Lasso need not line up in this way. Proposition 9 below shows what lining up means for coefficient vectors, and D2 and D3 compare the two cases.

M13 (p. 13). The paragraph after Proposition 8. "This proposition assesses what happens ... when the nuisance estimators both converge slowly" (after Proposition 7) is imprecise: one of them may converge fast; what matters is $b + c$. After Proposition 8, "in the proof of" starts a sentence in lower case, "27(ii)" needs "Lemma", and what uses $\hat{\Sigma}_1 = O_P(1)$ is Proposition 7(c). Replacement:

The proof is Lemma 26 in the appendix. Propositions 7 and 8 are two sides of the same calculation. In terms of the rates N^{-b} and N^{-c} of the two first steps, $b + c > 1/2$ suffices for a valid limit distribution, while $b + c < 1/2$ can make coverage go to zero. At $b + c = 1/2$ the bias is of the same order as the standard error, so coverage is distorted but need not go to zero.

Proposition 8 concerns the estimator. For the interval $\tilde{\theta} \pm z_{1-\alpha/2}(\hat{\Sigma}_1/N)^{1/2}$ to have asymptotic coverage $1-\alpha$, the variance estimator must also be consistent. In the proof of Chernozhukov et al. (2018, Theorem 3.2), the error of the variance estimator depends only on how fast each first step converges, not on the product of their errors, so consistency of $\hat{g}_k$ and $\hat{\ell}_{1k}$ is enough. I do not prove this here. Lemma 27(ii) shows the weaker statement $\hat{\Sigma}_1 = O_P(1)$, which is what Proposition 7(c) needs.

Proposition 8 is the result of Chernozhukov et al. (2018, Theorem 5.1) for the average treatment effect on the treated, applied to the change in outcomes: ψ_1 is their score (5.4) with Y replaced by $Y(1) - Y(0)$, except that θ enters as $-\theta$ instead of $-D\theta/p$. Their rate condition for this score involves only the product of the two errors. Chang instead uses their general condition (3.7)–(3.8), which bounds every error by one sequence $\varepsilon_N = o(N^{-1/4})$ and, in his Assumption 3.1, adds the squared propensity error. Lemma 4 shows that the squared

propensity error does not enter the bias of ψ_1, so the weaker condition suffices. The product condition guarantees validity in the limit; D2 shows that close to the boundary $b + c = 1/2$ convergence can be too slow to be visible at realistic sample sizes.

Note: This replaces the two paragraphs from "The proof is in the appendix. Proposition 7 and proposition 8 are two sides . . . " to "... bounds all rates by a single ε_N" (pp. 13–14).

2.3 Section 4.1: non-sparse data (pp. 14–15)

M14 (p. 14). The proof of the Remark has a wrong step (wrong as printed). "the numerator simplifies to $\mathrm{Cov}(g_0, \ell_{10}) = p_0/(1 - p_0)(\mathbb{E}[\ell_{10} \mid D = 0])$" is not true. Also make the Remark a numbered environment: "as per ." on p. 20 is an empty reference because the Remark has no number.

Proof. By Lemma 4, $B_1(p_0, c) = \mathbb{E}[(c-\ell_{10})(p_0-g_0)]/\{p_0(1-p_0)\}$. Since $\mathbb{E}[g_0] = p_0$, the constant c drops out and the numerator equals $\mathrm{Cov}(g_0, \ell_{10})$. Further, $\mathbb{E}[g_0\ell_{10}] = \mathbb{E}[D\ell_{10}] = p_0\mathbb{E}[\ell_{10} \mid D = 1]$ and $\mathbb{E}[\ell_{10}] = p_0\mathbb{E}[\ell_{10} \mid D = 1] + (1 - p_0)\mathbb{E}[\ell_{10} \mid D = 0]$, so $\mathrm{Cov}(g_0, \ell_{10}) = p_0(1 - p_0)\{\mathbb{E}[\ell_{10} \mid D = 1] - \mathbb{E}[\ell_{10} \mid D = 0]\}$. The last claim is Stein's lemma, $\mathbb{E}[Xf(X)] = \mathbb{E}[\nabla f(X)]$ for $X \sim N(0, I_p)$, applied to $f(X) = \Lambda(a + X'\gamma)$. □

M15 (pp. 14–15). Proposition 9: notation, and a simpler form.

- w is used both for the function $w(X)$ and for its mean, but $\mathbb{E}[Xw]$ and ∇w need the function.
- c_1 and c_2 are defined as expectations, but the proof writes "$= c_1\gamma+c_2\Delta\gamma$" for the expression inside the expectation.
- The third line of the proof has $\Lambda(a - X'\gamma)$; it is $\Lambda(a + X'\gamma)$.
- $\gamma\Delta\delta$ and $\Delta\gamma\Delta\delta$ are inner products and need transposes.
- c_1, c_2 are also the names of the adjustment terms in ψ_1, ψ_2 (p. 7) and of the three terms in Proposition 12.

While checking the proof I noticed that the two constants simplify, because $\Lambda' = \Lambda(1 - \Lambda)$:

$$\Lambda'(\hat{z})(1 - g_0) - \Lambda'(z)(1 - \hat{g}) = (1 - \hat{g})(1 - g_0)(\hat{g} - g_0), \qquad \frac{\Lambda'(\hat{z})(1 - g_0)}{(1 - \hat{g})^2} = \frac{\hat{g}(1 - g_0)}{1 - \hat{g}}.$$

With this, the constants can be read: $\bar{c}_2 = 1$ and $\bar{c}_1 = 0$ when $\hat{g} = g_0$, and the statement about "c_1 is small" becomes exact. I recommend this version (statement, proof and the text after it, replacing everything from Proposition 9 to "... can cancel in case not."):

Proposition (Bias in the Gaussian design)**.** *In this design,*

$$B_1(\hat{g}, \hat{\ell}) = \bar{c}_2\, \Delta\gamma'\Delta\delta + \bar{c}_1\, \gamma'\Delta\delta + (\hat{b} - b)\, \bar{w},$$

where, with $w(X) = (\hat{g} - g_0)/\{p_0(1 - \hat{g})\}$,

$$\bar{w} = \mathbb{E}[w(X)], \qquad \bar{c}_1 = \mathbb{E}\big[(1 - g_0)\, w(X)\big], \qquad \bar{c}_2 = \mathbb{E}\left[\frac{\hat{g}\,(1 - g_0)}{p_0(1 - \hat{g})}\right].$$

Proof. By Lemma 4 and $\hat{\ell} - \ell_{10} = (\hat{b} - b) + X'\Delta\delta$, $B_1 = \mathbb{E}[(\hat{\ell} - \ell_{10})\, w(X)] = (\hat{b} - b)\bar{w} + \Delta\delta'\mathbb{E}[X\, w(X)]$, and by Stein's lemma $\mathbb{E}[X\, w(X)] = \mathbb{E}[\nabla w(X)]$. Since $\hat{g} - g_0 = (1 - g_0) - (1 - \hat{g})$, we can write $w = p_0^{-1}\{(1 - g_0)/(1 - \hat{g}) - 1\}$. With $\nabla\hat{g} = \hat{g}(1 - \hat{g})\hat{\gamma}$ and $\nabla g_0 = g_0(1 - g_0)\gamma$,

$$\nabla w = \frac{(1 - g_0)\hat{g}(1 - \hat{g})\hat{\gamma} - g_0(1 - g_0)(1 - \hat{g})\gamma}{p_0(1 - \hat{g})^2} = \frac{(1 - g_0)\,(\hat{g}\,\hat{\gamma} - g_0\,\gamma)}{p_0(1 - \hat{g})}.$$

Substituting $\hat{\gamma} = \gamma + \Delta\gamma$ gives $\nabla w = \frac{(1-g_0)(\hat{g}-g_0)}{p_0(1-\hat{g})}\gamma + \frac{\hat{g}(1-g_0)}{p_0(1-\hat{g})}\Delta\gamma$, and taking expectations gives the result. □

The three terms have a clear reading. The first is the inner product of the two coefficient errors. The second and third are weighted averages of the signed propensity error $\hat{g} - g_0$; they vanish when the propensity score is correct, where $\bar{c}_2 = \mathbb{E}[g_0]/p_0 = 1$. If both first steps fit only intercepts, then $\Delta\gamma = -\gamma$, $\Delta\delta = -\delta$, $\bar{w} = 0$, $\bar{c}_2 = 1$ and $\bar{c}_1 = \mathrm{Var}(g_0)/\{p_0(1-p_0)\}$, and the proposition reproduces the Remark. When $\hat{g}$ is close to g_0, the bias is therefore approximately the inner product of the two coefficient errors,

$$B_1 \approx \bar{c}_2 \sum_{j=1}^{p} \Delta\gamma_j \, \Delta\delta_j. \tag{1}$$

For post-Lasso, the first step used in this thesis, the refit estimates the selected coefficients without shrinkage, so $\Delta\gamma_j \approx -\gamma_j \mathbf{1}\{j \notin \hat{S}_g\}$, $\Delta\delta_j \approx -\delta_j \mathbf{1}\{j \notin \hat{S}_\ell\}$ and

$$B_1 \approx \bar{c}_2 \sum_{j \notin \hat{S}_g \cup \hat{S}_\ell} \gamma_j \, \delta_j. \tag{2}$$

Only covariates that *both* fits omit contribute. This is the DML-DiD analogue of the double under-selection that Wüthrich and Zhu (2023) identify for post-double-selection. Without the refit, as in Chang's specification, products of the shrinkages of the selected coefficients are added. In DiD terms, γ_j measures how covariate j drives selection into treatment and δ_j how it drives the untreated trend. The bias comes from covariates that do both and that both fits miss. It is large when the products $\gamma_j\delta_j$ share a sign (*aligned* errors, as in Proposition 7(b)) and largely cancels otherwise. D2 and D3 test (1) and (2).

Note: Your code computes c_1 and c_2 in the old form. The two forms are algebraically identical, so the numbers in the tables do not change.

2.4 Section 4.2: the basis (pp. 15–19)

M16 (p. 16). Lemma 10. The statement ends "are in the same bases, q and $\tilde{q}$" and has $\hat{g}_K$; "hold the number of folds fixed" should be "hold the folds fixed"; the first line of the proof stops in the middle; the sentence about the Lasso belongs after the proof; and the lemma needs the intercept and the existence of the logit estimate. The caveats are needed in Section 6, where you drop a product group that has no controls.

Lemma (Invariance of unpenalised first steps)**.** *Let q_i contain an intercept, let $\tilde{q} = Aq$ with A invertible, and hold the folds fixed. Let $\hat{g}_k$ be the unpenalised logit fit and $\hat{\ell}_{1k}$ the OLS fit on the controls, both computed on I_k^c, and suppose the logit estimate exists. Then $\hat{g}_k$ and $\hat{\ell}_{1k}$, and hence $\tilde{\theta}$, are the same in the bases q and $\tilde{q}$.*

Proof. For every β, $q'\beta = \tilde{q}'\tilde{\beta}$ with $\tilde{\beta} = (A')^{-1}\beta$. The logit log-likelihood and the least-squares criterion depend on the coefficients only through the index $q'\beta$. Hence $\tilde{\beta}$ minimises the criterion in the basis $\tilde{q}$ exactly when $A'\tilde{\beta}$ minimises it in the basis q, and the fitted values coincide. □

The Lasso adds the penalty $\lambda \sum_j \hat{\psi}_j |\tilde{\beta}_j|$, which is not a function of the index alone, so the argument fails. The same argument shows that a Lasso fit is invariant to a change of basis that only involves columns whose coefficients are not penalised, for example a change of reference category among dummies that are forced into the model. Lemma 10 is the DiD counterpart of the observation of Kolesár et al. (2026) that OLS is invariant to the representation: with unpenalised first steps, $\mathcal{R}(\mathcal{A}) = 0$. There are two caveats. The logit estimate does not exist

when some category contains only treated or only untreated units, and when the number of columns is not small relative to the sample it is badly biased (Sur and Candès 2019). Outcome-regression DiD (OR-DiD), which needs only OLS among the controls, is invariant whenever the controls outnumber the columns of the dictionary and every category contains controls.

M17 (p. 16). Lemma 11. Correct. Two notational points: the difference of the two fits is called δ_g in the statement and δ in the bound and in the sentence after it, and δ is the vector of trend coefficients everywhere else. Call it Δg, as Proposition 12 does. "Let $g, g' < 1$" can simply be "Let $g, g' \in [\kappa, 1-\kappa]$" from the start.

M18 (pp. 16–17). Proposition 12 (wrong as printed: "=" for "≤").

- The bound is written with "=". It is an inequality, and the sum needs brackets around all three terms:

$$|\tilde{\theta}^{(A)} - \tilde{\theta}^{(A_0)}| \le \max_k \frac{1}{\hat{p}_k} \frac{1}{n} \sum_{i \in I_k} \Bigg(\|\Delta g_k\|_{P,\infty} \frac{(1-D_i)|\Delta Y_i - \hat{\ell}_k^{(A_0)}|}{\kappa^2} + \|\Delta g_k\|_{P,\infty} \frac{(1-D_i)|\Delta \ell_k|}{\kappa^2} + \frac{|\Delta \ell_k|}{\kappa} \Bigg).$$

- The title "Estimator bias in response to basis shift" and "the deviance in the estimators" and "The bound of the bias": nothing here is a bias. It is the *difference between two estimates*. Title: "Change of the estimate under a change of basis".
- "$\Delta \ell_k \equiv \hat{\ell}_j^{(A)} - \hat{\ell}_j^{A_0}$, $j = \{1, 2\}$" should be $\Delta \ell_k \equiv \hat{\ell}_{1k}^{(A)} - \hat{\ell}_{1k}^{(A_0)}$.
- The three terms are named c_1, c_2, c_3; see M15.
- In $\mathcal{R}(\mathcal{A})$, say which variance: "$\hat{\Sigma}_1^{(A_0)}$, the variance estimate in the reference basis A_0".
- "If the rate conditions hold uniformly over a fixed and finite set of bases, the first order terms in (12) vanish, and $\mathcal{R}(\mathcal{A}) = o_p(1)$": the conclusion is right, but it does not follow from Proposition 12. It follows from Lemma 26. Replacement:

If the conditions of Proposition 8 hold in every basis of a fixed, finite set $\mathcal{A}$, then by Lemma 26 every $\tilde{\theta}^{(A)}$ has the same expansion $N^{-1} \sum_i \varphi(W_i) + o_P(N^{-1/2})$, which does not depend on the basis. The estimates then differ by $o_P(N^{-1/2})$, and $\mathcal{R}(\mathcal{A}) = o_P(1)$. A range that is not small relative to one standard error is therefore evidence that the rate conditions, and in this context sparsity, fail in at least one of the bases.

M19 (p. 17). "INDSÆT HER". The running example goes here; the text and the table are in Part 4. "To formalise this insight, consider the following lemma" introduces a proposition.

M20 (p. 17–18). Proposition 13 (wrong as printed).

- The sum over the pools is missing in the display. As printed, B_1 is the contribution of a single, unspecified pool G.
- "suppose both full and threshold coding, and that" is not a sentence. It should be "suppose, as holds for reference and threshold coding, that".
- "if their rows are similar" should be "if their rows agree".
- The proof has a sign error: $\mathbb{E}[(\hat{\ell} - \ell_{10})(\hat{g}_G - g_0) \mid Z \in G]$ equals $\mathrm{Cov}(\ell_{10} - \hat{\ell}, g_0 \mid Z \in G)$, not $\mathrm{Cov}(\hat{\ell} - \ell_{10}, g_0 \mid Z \in G)$. (The statement has the right sign.)

- "with a reference category spans all functions of Z": it is the intercept together with the columns that spans them.

The corrected statement and proof are in Part 4.

M21 (p. 18). Corollary 14 and the two remarks. The corollary and its proof are correct ("If it atypical in both" needs "is"). Remark 1 ends with a wrong conclusion **(wrong as printed)**: "An unlucky choice of category therefore matters when it is large, i.e. when it comprises a majority of the observations." The condition you derive in the sentence before is that $\pi_b\sqrt{N}$ is not small. In D4a, a reference category with 5% of the sample gives coverage 0.77 at $N = 4000$, and one with 15% gives 0.06. Corrected remarks are in Part 4.

2.5 Appendix (pp. 28–44)

M22 (pp. 28–29). Chebyshev. Definition 18 has $t\sigma_X^2$; it is $t\sigma_X$. In the proof of Lemma 19, "Since both $\Pr(|X - \mathbb{E}(X)| \geq t)$ and $\mathrm{Var}(X)/t^2$ are positive" is not the reason the two events are equal. The reason is that $t > 0$ and $x \mapsto x^2$ is increasing on $[0, \infty)$. (I recommend cutting this material altogether, see Part 9.)

M23 (pp. 33–35). Lemma 25. The decomposition and its proof are correct. Fix: "Lemma (4)" twice (it prints like an equation number); "$\hat{p}_k$ is the mean of $N - n$ Bernoulli(p_0) variables" should be n (M4); and before "On the event that $\{\hat{p}_k \geq p_0/2\}$" add "whose probability tends to one, since $\hat{p}_k \xrightarrow{p} p_0 > 0$". The step $\mathbb{E}[\psi_1^2] \leq \mathbb{E}[(\psi_1 + \theta_0)^2]$ is true only because $\mathbb{E}[\psi_1] = 0$; say so.

M24 (p. 35). Lemma 26.

- The title is "The variance estimator"; the lemma is asymptotic linearity.
- "Fix number of folds k" should be "Fix a fold k". In (i) a bracket is missing. In (ii) "On the event that $\Pr(\{\kappa' \leq \hat{g}_k \leq 1 - \kappa'\}) \to 1$" should be "On the event $\{\kappa' \leq \hat{g}_k \leq 1 - \kappa'\}$, whose probability tends to one".
- Step (v) is garbled, and the sentence after the display argues with auxiliary samples, which you no longer use for $\hat{p}_k$.

Replacement for (v) and the rest of the proof:

> (v) Since $p_0/\hat{p}_k - 1 = -(\hat{p}_k - p_0)/p_0 + (\hat{p}_k - p_0)^2/(p_0\hat{p}_k) = -(\hat{p}_k - p_0)/p_0 + O_P(N^{-1})$ and $\hat{p}_k - p_0 = \mathbb{E}_{n,k}[D - p_0]$,
>
> $$\tilde{\theta}_k - \theta_0 = \mathbb{E}_{n,k}\Big[\psi_1(W, \theta_0, p_0, \eta_{10}) - \frac{\theta_0}{p_0}(D - p_0)\Big] + o_P(N^{-1/2}) = \mathbb{E}_{n,k}[\varphi(W)] + o_P(N^{-1/2}).$$
>
> The folds partition the sample and have the same size $n = N/K$, so averaging over k gives $\tilde{\theta} - \theta_0 = N^{-1}\sum_{i=1}^N \varphi(W_i) + o_P(N^{-1/2})$. φ has mean zero, since $\mathbb{E}[\psi_1(W, \theta_0, p_0, \eta_{10})] = 0$ and $\mathbb{E}[D] = p_0$. Its variance is finite by overlap and Assumption 3(d), and positive by Assumption 3(e). The central limit theorem completes the proof.

M25 (pp. 36–37). Lemma 27. The proof is correct. Fix:

- Statement: "the set $\Pr(A_k(\kappa'))) \to 1$" should be "$\Pr(A_k(\kappa')) \to 1$", and $\delta_{g,k} \equiv \hat{g}_k - g_0$, $\delta_{\ell,k} \equiv \hat{\ell}_{1k} - \ell_{10}$ must be defined in the statement. There is a stray ". :" before (i).
- First display of the proof: both terms should be multiplied by p_0, i.e. "$-p_0\theta_0$" twice (they cancel, so nothing else changes).
- "by using lemma ??" is Lemma 11.

- In the display for $\mathbb{E}[(\Delta\psi)^2]$, the factor $1/p_0$ belongs to both terms and the second term has a minus sign: $\Delta\psi = p_0^{-1}\{(w_g - w_0)V_1 - w_g\delta_\ell\}$. The bound in the next step is unaffected. The line with the absolute value around the whole expression is redundant.
- "On the set $A_k(\kappa')$, which per construction depends on the auxiliary sample I_k^c, the On $A_k(\kappa')$, which depends only on I_k^c": delete the first half. "since we assumed that $\|\delta_g\|, \|\delta_g\| = o_P(1)$": the second is δ_ℓ.
- (ii): "Using the variance form in lemma 26" should be "Using the definition of $\hat{\Sigma}_1$". "$4/(\hat{p}_k^2\kappa'^2)$ and $2\tilde{\theta}$" should be $2\tilde{\theta}^2$. The citation "(vandervaart)" for "it is enough to show that the variance within each fold is bounded" is not needed: a finite sum of $O_P(1)$ terms is $O_P(1)$.

M26 (pp. 38–41). Proof of Proposition 7. Correct. Fix:

- Part (a) starts "By assumption, $P(A_k(\kappa')) \to 1$". The assumption of (a) is $A_k(\kappa)$.
- "in order to transfer it to condition 3.1(d) in Chang": the band is part of condition (f) in Chang's Assumption 3.1, i.e. Assumption 3(f)(ii) in your numbering; (d) is the bound on $\mathbb{E}[V_1^2 \mid X]$.
- "Where I have used that $0 \leq 1 - g_N \leq 1$": you divide by $1 - g_N$, so $0 < 1 - g_N \leq 1$.
- "$B_1(g_N, \ell_N) = O(N^{-(b+c)}) = o(1)$": you have shown more, namely that $B_1(g_N, \ell_N)$ is between $N^{-(b+c)}\mathbb{E}[h^2]/p_0$ and $2N^{-(b+c)}\mathbb{E}[h^2]/(p_0\kappa)$. Say that; it is what (b) claims.
- The display for the average over folds lacks brackets and the term $(p_0/\hat{p}_k - 1)\theta_0$; write "$\tilde{\theta} - \theta_0 = B_1(g_N, \ell_N) + O_P(N^{-1/2})$ by averaging the previous display over k".
- Part (c): T_N is first defined as an inequality ("$T_N = \sqrt{N}|\tilde{\theta} - \theta_0| \leq \hat{\sigma}z$"). Define the statistic $T_N \equiv \sqrt{N}|\tilde{\theta} - \theta_0|$ and the coverage event $\{T_N \leq z\hat{\sigma}\}$ separately.
- "The set $T_N \leq M \cap \{|(N^{-(b+c)}\mathbb{E}[h^2/p_0])(1-\delta)|\}$ is only non-empty when ..." is not a set. Write: "On $\{|\rho/B_1(g_N, \ell_N)| \leq \delta\}$ we have $T_N \geq \sqrt{N}N^{-(b+c)}\mathbb{E}[h^2/p_0](1-\delta)$, so $\{T_N \leq M\}$ can only occur if this lower bound is at most M or if $|\rho/B_1(g_N, \ell_N)| > \delta$."
- δ is again a new object here; use ϵ' or τ.

> The argument carries over to repeated cross-sections with ψ_2, ℓ_{20} and V_2, and with the version of Lemma 4 that has the factor $1/\{\lambda_0(1-\lambda_0)\}$. One term is new. Unlike $\hat{p}_k$, which multiplies the whole fold average, $\hat{\lambda}_k$ enters the score itself, and the score is not orthogonal in λ. Conditional on I_k^c, $\hat{\lambda}_k$ is fixed and shifts the conditional mean of the score by $G_{2\lambda_0}(\hat{\lambda}_k - \lambda_0) + O_P(N^{-1}) = O_P(N^{-1/2})$, because $\hat{\lambda}_k$ is the mean of $N - n$ Bernoulli(λ_0) variables. It also moves the target of the outcome Lasso, $\mathbb{E}[(T - \hat{\lambda}_k)Y \mid X, D = 0]$, away from ℓ_{20} by $O_P(N^{-1/2})$. Since $b + c < 1/2$, both are dominated by the bias of order $N^{-(b+c)}$, and parts (a)–(c) hold as stated. Under the conditions of Proposition 8, the first of the two terms is of the same order as the leading term; it is the reason for the term $\hat{G}_{2\lambda}(T - \hat{\lambda}_k)$ in $\hat{\Sigma}_2$.

M27 (p. 41). The paragraph on repeated cross-sections stops without a conclusion.

M28 (pp. 42–44). Remaining. "On the $[\kappa, 1.\kappa]$ bound" should be $[\kappa, 1-\kappa]$; "using lemma ??" is Lemma 11; in the regularity conditions for repeated cross-sections, (a) should be $\Pr(\kappa \leq g_0(X) \leq 1-\kappa) = 1$, "$D = g_o(X) + U$" should be g_0, and the wording problems of M6 are repeated in (a), (b), (g) and (h).

3 Inserts: title, introduction and Section 2

3.1 Title and section titles

The title still reads "My advisor told me sparsi ty was an assumption and he was right", with a broken word; Section 5 is "Monte Carlo shows I'm right" and Section 6 "Replication of a replication empirical application (say it fast twice)". Unless your supervisor has agreed to these, replace them. Suggestions:

- Title: *Sparsity is a choice: representation and bias in double machine learning for difference-in-differences*. Plainer: *Double machine learning for difference-in-differences when sparsity fails*.
- Section 4: *Sparsity, representation and bias* (now "Enter sparsity").
- Section 5: *Simulation evidence*.
- Section 6: *Chang's application revisited*.
- Section 7: *Conclusion*.

3.2 Introduction (pp. 1–3)

Where: p. 1, first sentence.

The comparison of the two London water companies before and after one of them moved its intake, which is the usual reference for the first difference-in-differences, is in the second edition of Snow's book, from 1855. The 1849 edition is a short pamphlet without it. Write "Ever since John Snow's study of cholera in London (Snow 1855), . . . ".

Where: p. 1, second paragraph.

"which is why several add-ons exist" is informal ("several extensions exist"), and "heterogeneous treatment intensity, depending on the characteristics of the treated unit" does not describe Callaway et al. (2024), which is about a treatment that varies in intensity (dose) across units. Write "or allow the treatment to vary in intensity across units".

Where: p. 2: "He shows that the resulting estimator, the DML-DiD, is valid whenever the first-step estimators converge faster than $N^{-1/4}$."

> He shows that the resulting estimator, DML-DiD, is $\sqrt{N}$-consistent and asymptotically normal when both first-step estimators converge faster than $N^{-1/4}$.

Where: p. 2: "**kolesar<empty citation>** show that sparsity is . . . " and the next paragraph.

The citation key is broken here (and for Abadie on pp. 6–7, Chang, Sequeira and Kolesár on pp. 24–25, Chernozhukov on p. 37, Pitman on pp. 28–29, van der Vaart on pp. 29 and 37). Search the source for `{kolesar}`, `{abadie}`, `{chang}`, `{sequeira}`, `{chernozhukov}`, `{pitman}` and `{vandervaart}` and insert the full keys. In the same paragraph: "sparsity-based ML method" needs a plural, "choices that does not affect" should be "do not", and "where the researcher choice enter" should be "where the researcher's choices enter".

Where: p. 3, top: "a representation that makes only one nuisances can be harmless . . . in the untreated change in outcomes."

A word is missing in the second feature and the third is hard to follow.

> Second, the bias depends on the *product* of the two first-step errors: a representation that makes only one nuisance non-sparse can be harmless, while one that makes both non-sparse in the same direction is not. Third, the outcome nuisance is a trend rather than a level, so what has to be sparse is how the covariates shift the untreated *change* in outcomes.

Where: p. 3: the paragraph "The thesis makes three contributions . . . " and the outline after it.

Contribution (a) claims as new what is known (Part 9), contribution (b) leaves out the pooling result, and the outline is off by one section and promises a conclusion in "section 6".

> The thesis makes three contributions. (a) It states the conditions for DML-DiD in terms of the product of the two first-step errors. That the product is what matters is known for doubly robust estimators (Chernozhukov et al. 2018; Smucler et al. 2019); I show that it replaces Chang's requirement on each first step, and that coverage can go to zero when the product condition fails. (b) It derives the bias that a failure of sparsity creates. In a Gaussian design, the bias is the inner product of the two first-step coefficient errors. With categorical covariates and post-Lasso first steps, it is the covariance of selection and trends within the groups of categories that the Lasso pools. This makes precise when the choice of reference category matters: when the reference category differs from the categories it is pooled with in *both* selection and trends, in proportion to its share of the sample. (c) It documents these mechanisms by simulation, and revisits Chang's empirical application to measure how much choices of representation move the estimate.
>
> The rest of the thesis is organised as follows. Section 2 places the thesis in the literature. Section 3 reviews the estimators of Abadie and Chang. Section 4 contains the theory: what happens to DML-DiD when sparsity fails, and how this depends on the representation of the covariates. Section 5 provides simulation evidence. Section 6 revisits Chang's application. Section 7 concludes.

Note: If Section 6 ends up short or is cut, change (c) accordingly.

3.3 Section 2 (pp. 3–5)

Where: p. 3: the linear model.

The model $Y_i = D_i\theta + X_i'\beta + \varepsilon_i$, $D_i = X_i'\gamma + v_i$ is the setting of post-double-selection, and of Kolesár et al. Say so, since the reader otherwise wonders why a section on DiD starts with a cross-section: "... consider the partially linear model that Belloni et al. (2014) and Kolesár et al. (2026) study". "Suppose that as a researcher you are interested in modeling the outcome Y_i" should be "interested in θ".

Where: p. 3–4: "Goodman-Bacon (2021) and Caetano and Callaway (2026) document naïve linear inclusion . . . as e.g. formulated by Angrist (n.d.)".

A word is missing ("document *that* naïve linear inclusion"), the book is Angrist and Pischke (2009), and "diverges from non-staggered treatment timing with homogenous effects" packs two different problems into one clause: Goodman-Bacon is about timing, Caetano and Callaway about covariates. Split the sentence in two.

Where: p. 4: "Chang (2020) shows that plugging machine learning estimates . . . destroys $\sqrt{N}$-consistency. He restores it by deriving new score functions of the outcome- and propensity regression . . . "

> Chang (2020) shows that plugging machine learning estimates of the propensity score into the estimator of Abadie (2005) does not in general give a $\sqrt{N}$-consistent estimator. He restores $\sqrt{N}$-consistency by deriving Neyman-orthogonal scores, which add an outcome regression to Abadie's moment condition, and by combining them with cross-fitting.

Where: p. 4: "Ordinary Least Squares becomes infeasible, while sparsity-based estimators remain valid."

Too strong: their validity is what the thesis questions. Write "... OLS becomes infeasible or very imprecise, while sparsity-based estimators remain feasible".

Where: p. 5: "Smucler et al. (2019) show that ... the final estimator will be $\sqrt{n}$-consistent."

"two-nuisance influence function" is their "bilinear influence function", and the last sentence says the estimator is consistent "as long as the other nuisance [is] estimated consistently", which is not enough: the other one must be sparse enough to compensate.

> Smucler et al. (2019) show that for parameters with a bilinear influence function, which include the ATT, cross-fitted estimators with ℓ_1-regularised first steps are $\sqrt{n}$-consistent and asymptotically normal whenever the product of the sparsity indices of the two nuisances is small enough, instead of each index on its own. The authors call this feature rate double robustness: one nuisance may be far from sparse, provided it is still estimated consistently and the other is sparse enough to compensate.

Where: p. 5: Wüthrich and Zhu.

"under half the chosen penalty" is right. Add what makes it relevant here: "...can lead to bias in post-double-selection. Proposition 9 gives the corresponding result for DML-DiD."

4 Inserts: categorical covariates and pooling (Section 4.2)

Where: p. 17, from "Proposition 12 shows how far the estimator moves ..." to the end of Section 4 (p. 19). This replaces "INDSÆT HER", Proposition 13, its proof, Corollary 14 and the two remarks.

Note: I recomputed every number in the running example and in Table 2 today from `Bribes_Regression.dta` in the repository, with the sample definition of your notebook ($N = 1084$). Recompute them once in your own notebook (one `groupby` each), so that you can answer questions about them. The product group variable is `hc_group`; transportation is group 14, machinery 13, animal products 1 and metals 12.

> **Categorical covariates: what the Lasso pools**
>
> Proposition 12 describes how the estimate moves between bases. With categorical covariates it is possible to say *which* bases create bias. As a running example, take the product group of a shipment in the data of Sequeira (2016) that Chang uses. The shipments fall into 14 product groups. The largest, transportation (vehicles), accounts for 68.5% of the 1,084 shipments in Chang's sample, and it differs from the other groups in both selection and trends: 5.1% of its shipments are treated, against 40.4% in the other groups, and among untreated shipments the mean of $\log(1 + \text{bribe})$ is flat for transportation (0.91 before 2008, 0.88 after), while it falls from 7.95 to 1.01 in the other groups. The comparison rests on 60 and 24 untreated shipments before 2008 and is only descriptive. It shows, however, that the way product group enters the dictionary can matter.
>
> A post-Lasso fit on dummies gives each category with a selected dummy its own level, like a fixed effect, and gives all remaining categories one common level, like pooled OLS. I say that these categories are *pooled*. Just as pooled OLS is biased when the ignored group effects are correlated with the regressor, pooling biases DML-DiD when, among the pooled categories, those more often treated also have different untreated trends. Table 2 illustrates how the reference category decides which categories are pooled.

> To formalise this, let $Z \in \{1, \dots, C\}$ be a categorical covariate with shares $\pi_c = \Pr(Z = c) > 0$, and suppose, to isolate its role, that g_0 and ℓ_{10} depend on X only through Z. Any coding of Z by 0/1 columns together with an intercept spans all functions of Z if it has full rank. A post-Lasso fit on the selected columns is constant on the groups of categories that the selected columns do not distinguish: two categories are pooled if their rows agree in every

Product group	Share	Treated	Reference: transportation				Reference: metals			
			d^*_{mach}	d_{anim}	d_{met}	Pool	d^*_{trans}	d_{mach}	d_{anim}	Pool
Transportation	0.685	0.051	0	0	0	I	1	0	0	II
Machinery and electrical	0.072	0.667	1	0	0	II	0	1	0	I
Animal products	0.051	0.018	0	1	0	I	0	0	1	I
Metals	0.042	0.422	0	0	1	I	0	0	0	I

Table 2: The four largest of the 14 product groups in Chang's sample of $N = 1084$ shipments: share of shipments and share treated, and two codings by dummies with different reference groups. The starred column is the one the Lasso is assumed to select. Two groups are pooled if their rows agree in every selected column; post-Lasso then gives them the same fitted propensity score and outcome trend. With transportation as the reference, it is pooled with every group whose dummy is not selected. With metals as the reference, transportation has its own dummy and, once that dummy is selected, its own level. Source: own calculations based on the replication data of Chang (2020); the names of the groups are inferred from the HS codes in each group.

selected column. Let $\mathcal{G}_g$ and $\mathcal{G}_\ell$ be the resulting partitions of the categories for the two fits, and suppose, as holds for reference and threshold coding, that the selected columns and the intercept span all functions that are constant on these pools. In the population, the refitted propensity score in pool G is then the share of treated in the pool, $\hat{g}_G = \mathbb{E}[g_0 \mid Z \in G]$.

Proposition (Pooling bias)**.** *Under these conditions,*

$$B_1(\hat{g}, \hat{\ell}) = \sum_{G \in \mathcal{G}_g} \frac{\Pr(Z \in G)}{p_0(1-\hat{g}_G)} \operatorname{Cov}\left(g_0,\, \ell_{10} - \hat{\ell} \mid Z \in G\right).$$

If both fits pool the same categories, $\hat{\ell}$ is constant on each pool and the covariance is $\operatorname{Cov}(g_0, \ell_{10} \mid Z \in G)$*.*

Proof. On a pool G, $\hat{g} = \hat{g}_G$ is constant and $\mathbb{E}[\hat{g}_G - g_0 \mid Z \in G] = 0$. Hence $\mathbb{E}[(\hat{\ell} - \ell_{10})(\hat{g}_G - g_0) \mid Z \in G] = \operatorname{Cov}(\ell_{10} - \hat{\ell},\, g_0 \mid Z \in G)$, and $1 - \hat{g}$ is constant on G. Summing over the pools with weights $\Pr(Z \in G)$ in Lemma 4 gives the result. If $\hat{\ell}$ is constant on G, it drops out of the covariance. □

Proposition 13 is the Remark of Section 4.1 applied pool by pool. When both fits pool the same categories, each pool contributes the bias of an unadjusted DiD within the pool, $\operatorname{Cov}(g_0, \ell_{10} \mid Z \in G)/\{\hat{g}_G(1-\hat{g}_G)\}$, weighted by the pool's share of the treated, $\Pr(Z \in G \mid D = 1) = \Pr(Z \in G)\,\hat{g}_G/p_0$. Pooling is harmless where selection or the remaining trend is constant within the pool, and harmful where both vary together. With a single pool, the proposition reduces to the Remark.

Corollary (Reference category)**.** *Use reference coding with reference category b, and suppose that in both fits the pool containing b is $\{b\} \cup U$, where the categories in U share a common level: $g_c = g_U$ and $\ell_{10,c} = \ell_U$ for $c \in U$. Then, with $w = \pi_b/(\pi_b + \pi_U)$, the pool $\{b\} \cup U$ contributes*

$$\frac{\pi_b(1-w)}{p_0(1-\hat{g}_{\{b\}\cup U})}\,(g_b - g_U)(\ell_{10,b} - \ell_U)$$

to B_1. The contribution is zero if the reference category is typical in selection ($g_b = g_U$) or in trends ($\ell_{10,b} = \ell_U$). It is non-zero only if the reference category is atypical in both, its sign is the sign of $(g_b - g_U)(\ell_{10,b} - \ell_U)$, and it grows with the share π_b of the reference category.

Proof. Within the pool, g_0 and ℓ_{10} take two values, with probability w on b, so $\operatorname{Cov}(g_0, \ell_{10} \mid Z \in \{b\} \cup U) = w(1-w)(g_b - g_U)(\ell_{10,b} - \ell_U)$. Multiply by $\Pr(Z \in \{b\} \cup U) = \pi_b/w$. □

The pool $\{b\} \cup U$ arises when neither fit selects the dummies of U, and this is the regime the Lasso is pushed into. With an atypical reference category, every typical category needs the same non-zero coefficient. It is cheaper for the Lasso to fit the common level through the unpenalised intercept and leave these coefficients at zero, which pools the reference category with the typical ones. Two remarks follow.

- *Size.* In the regime of Kolesár et al. (2026), with many categories of equal size, π_b is of order $1/p$, so a wrong reference category costs only $O(1/p)$ in approximation error and satisfies their approximate sparsity condition (7). In DML-DiD, the bias is of order π_b, and it is negligible relative to the standard error only if $\pi_b\sqrt{N} \to 0$. A wrong reference category therefore matters as soon as its share is not small relative to $N^{-1/2}$. This is the typical case in applied work, where a common default is to use the largest category as the reference.
- *Ordered categories.* Suppose Z is ordered, for example in bins, and both selection and trends jump at the same bin. With dummy coding, the Lasso must select a separate dummy for every bin on one side of the jump, and each dummy rests only on the observations in its own bin. With threshold coding, $\mathbf{1}\{Z \geq c\}$, a single column describes the jump, and it rests on all observations. When the bins are too small for their dummies to be selected, dummy coding pools bins on both sides of the jump, and by Proposition 13 the bias approaches that of the unadjusted DiD. Threshold coding avoids this.

5 Inserts: Section 5, simulations

All numbers in this section are computed from the CSV files on the branch (`d1_*.csv`, `d23_*.csv`, `d4_*.csv` and `d4_*_forced.csv`). The tables are in `tables/`; the scripts `tables/build_d23.py` and `tables/build_d4.py` rebuild them. They need `\usepackage{booktabs}`.

5.1 Practicalities and estimators (pp. 19–20)

What to fix.

- n is the fold size N/K everywhere else, but here it is the number of observations in the Lasso. Use M, as in Section 3.3.
- "The error $\hat{e} = y - \bar{y}$ is calculated in a pilot step and the pilot residuals in one update" is hard to parse.
- "I follow the code of Chang (2020) and clip": Chang's code does not clip. It drops the observations with $\hat{g}$ outside $(0.03, 0.97)$. Your runs clip, so only the wording changes.
- The list of estimators does not match the tables. Table 1 has two versions of DML-DiD and Abadie's estimator, which the list lacks; OR-DiD appears only in D4; and $K = 5$ is not true for the first panel of Table 1.
- "Each design cell is replicated $R = 500$ times": D4 has $R = 200$. "replication r uses the r-th child of that seed": no seed has been introduced.
- "the product term of Lemma ??" (twice, also in the caption of Table 1): Lemma 4.

Where: Replace "Practicalities" and "Estimators" (p. 19 to the top of p. 20).

First steps. Covariates are standardised on each training sample, the intercept is never penalised, and all penalties are plug-in with $c = 1.1$ and $\alpha = 0.05$. Let M be the number of observations in the regression at hand: the controls of the auxiliary sample for the outcome

nuisance, and the whole auxiliary sample for the propensity score. The outcome nuisance solves

$$\min_{\beta} \ \frac{1}{M}\sum_i (y_i - q_i'\beta)^2 + \lambda \sum_j \hat{\psi}_j |\beta_j|, \qquad \lambda = \frac{2c}{\sqrt{M}}\,\Phi^{-1}\Big(1-\frac{\alpha}{2p}\Big), \quad \hat{\psi}_j = \Big(\frac{1}{M}\sum_i q_{ij}^2 \hat{e}_i^2\Big)^{1/2}.$$

The residuals $\hat{e}_i$ are first $y_i - \bar{y}$ and then, in one refinement step, the residuals of the resulting Lasso fit. This is the rule of Belloni et al. (2012) that Chang suggests, with $\alpha = 0.05$ in place of a level that tends to zero. The propensity score solves

$$\min_{\beta} \ \frac{1}{M}\sum_i \left[\log(1+e^{q_i'\beta}) - D_i q_i'\beta\right] + \lambda_g \|\beta\|_1, \qquad \lambda_g = \frac{c}{2\sqrt{M}}\,\Phi^{-1}\Big(1-\frac{\alpha}{2p}\Big),$$

a plug-in penalty of the form proposed by Belloni et al. (2016). I clip the fitted propensity scores to $[0.03, 0.97]$. This is the window Chang's replication code uses; his code drops the observations outside it instead.

Estimators.

1. *DML-DiD, post-Lasso*: the cross-fitted estimator of Definition 2 with $K = 5$ folds, where each nuisance is refitted without penalty on its selected covariates. This is the main specification.
2. *DML-DiD, Lasso (no refit)*: the same, with the penalised fits plugged into the score directly, as in Chang's specification.
3. *Oracle*: the score evaluated at the true (g_0, ℓ_{10}). Its error is sampling error alone.
4. *Unadjusted DiD*: the difference in mean outcome changes between treated and controls.
5. *Abadie, plug-in Lasso* (D1 only): the estimator (3) with $\hat{g}$ from a logit Lasso fitted on the full sample, without refit and without cross-fitting.
6. *OR-DiD* (D4 only): OLS of $Y(1) - Y(0)$ on the full dictionary among the controls, imputed for the treated. It requires fewer regressors than controls.

Replications and reporting. D1 and D2/D3 use $R = 500$ replications per design cell and D4 uses $R = 200$. All random numbers derive from one seed: in each cell, replication r uses the r-th child of a seed sequence built from that seed and the parameters of the cell, so cells with the same data-generating process share their data. For each estimator I report the bias, the standard deviation of $\tilde{\theta}$ across replications (SD), the coverage of the nominal 95% interval $\tilde{\theta} \pm 1.96\,\widehat{\mathrm{SE}}$, and the ratio of the average standard error to the SD (SE/SD). Monte Carlo standard errors (MCSE) measure the simulation uncertainty of a reported number, which shrinks with R; the SD measures the sampling variability of the estimator itself. For a coverage rate c the MCSE is $\sqrt{c(1-c)/R}$, about 0.010 at nominal coverage for $R = 500$ and 0.015 for $R = 200$. For DML-DiD I also report B_1, the product term of Lemma 4, evaluated with the fold-specific fits on an independent test draw of 5,000 observations and averaged over folds.

Each design tests specific results of Section 4. D1 tests Lemma 4 and the Remark of Section 4.1 in a sparse design. D2 and D3 test Propositions 7–9 when the coefficients are dense. D4 tests Lemma 10, Proposition 13 and Corollary 14 when the truth is sparse in one coding but not in another.

5.2 D1 (pp. 20–22)

What to fix.

- **Table 1 is from the run before your rerun.** The file `d1_table.tex` in the repository is

built from the current CSV files and differs in the third decimal in ten entries (for example Abadie $1.428 \to 1.429$, $1.089 \to 1.090$, $0.944 \to 0.945$; robust SD of post-Lasso at $N = 200$ $0.530 \to 0.524$; coverage of Lasso in the first panel $0.946 \to 0.948$). Nothing changes in substance, but paste the current file so that the thesis matches the files you hand in.

- "before any notion of sparsity is introduced": the design is exactly sparse. Write "in a design where sparsity holds".
- "when parallel trends do not hold unconditionally for the untreated units" should be "when parallel trends hold only conditionally on the covariates".
- "$\ell_{10} = 1 + X'\gamma$ with $\gamma = \delta$" should be $\ell_{10} = 1 + X'\delta$ with $\delta = \gamma$.
- "the same five covariates with the same weights will drive the trend for the treated and the untreated" is not what the design does. The same five covariates drive *selection into treatment* and *the untreated trend*.
- "I a signal-to-noise-ratio such that it explains half the variation in the untreated outcome": a verb is missing, and it is the *change* in the outcome.
- "as per .": the Remark.
- Results: "it shows that the plug-in Abadie estimator is skewed" should be "biased". "between $1.563 - 1.569$": the table has 1.559–1.566. "removes most of the bias: at $N = 200$": at $N = 200$ a third of it remains. "the refitting removes some of the confounding" says too little: without the refit coverage is zero in every trend cell.
- Digest: "requieres N to be around 1000 for close to full coverage" **(wrong as printed)**: coverage is 0.884 at $N = 1000$ and reaches the nominal level only at $N = 2000$. "refitting the penalty": the penalty is not refitted; the coefficients are refitted after selection.
- The digest says "D1 shows that the bias of DML-DiD is the product term of lemma 4", but the results text never compares the bias with B_1.

Where: Replace "Design setup" from "I deviate from his setup . . . " to ". . . while maintaining letting $p = N/2$." (p. 20), and add the first paragraph after the display for $Y(1) - Y(0)$.

> The level coefficients β cancel in the difference. Whether the outcome *levels* are sparse is therefore irrelevant for DML-DiD; in the code, $\beta_j = \gamma_j + 0.5$, so every covariate shifts the level. Only the untreated trend ℓ_{10} and the propensity score matter. Following Chang (2020), the first version of the design has $\ell_{10} \equiv 1$, $N = 200$, $p = 100$, $K = 2$, and errors with standard deviation 0.1.
>
> I depart from his design in three ways. First, I use $K = 5$ folds. Second, I add a version with a trend, $\ell_{10}(X) = 1 + X'\delta$ with $\delta = \gamma$: the five covariates that drive selection into treatment also drive the untreated trend, so parallel trends hold conditionally on X but not unconditionally. In this version the standard deviation of ε_2 is $\|\delta\| = \sqrt{2.2}$, so that in the population the covariates explain half of the variance of the untreated change in outcomes. Third, I let $N \in \{200, 1000, 2000\}$ with $p = N/2$. With this γ, the propensity index has standard deviation 1.48, so overlap is thin in the tails: some controls receive weights $\hat{g}/(1 - \hat{g})$ above 30, which inflates the variance of the estimators that use the propensity score.

Note: "above 30" is the clipping bound, $0.97/0.03 = 32$. Keep the sentence only if the clipping binds in your D1 runs (check `np.mean(g == 0.97)` once).

Where: Replace "Priors" (pp. 20–21).

Priors. In Chang's design the untreated trend is constant, so $\mathrm{Cov}(g_0, \ell_{10}) = 0$ and, by the Remark of Section 4.1, the unadjusted DiD is unbiased: there is no confounding to remove. DML-DiD should be unbiased as well. Abadie's estimator is the score ψ_1 with $\ell \equiv 0$, so by Lemma 4 its bias for a given propensity fit is $B_1(\hat{g}, 0) = \mathbb{E}\big[\ell_{10}\,(g_0 - \hat{g})/\{p_0(1-\hat{g})\}\big]$. In Chang's design, this is the common trend, $\ell_{10} \equiv 1$, times the average error in the weights of the controls, which is not zero when $\hat{g}$ is shrunk. I therefore expect Abadie's estimator to be biased, by an amount proportional to the common trend, although nothing confounds the comparison.

With the trend, the unadjusted DiD is biased by

$$\frac{\mathrm{Cov}(g_0, \ell_{10})}{p_0(1-p_0)} = \frac{\mathbb{E}[\Lambda'(X'\gamma)]\,\gamma'\delta}{p_0(1-p_0)} = 1.563,$$

since $p_0 = 0.5$, $\gamma'\delta = 2.2$ and $\mathbb{E}[\Lambda'(X'\gamma)] = 0.178$. DML-DiD is consistent, but at small N the smallest coefficients lie below the plug-in penalty and the Lasso sets them to zero. Lemma 4 then predicts a finite-sample bias equal to B_1, which falls as N grows, and Proposition 9 predicts that without the refit the shrinkage of the selected coefficients adds to it.

Where: Replace the two paragraphs of results after Table 1 (pp. 21–22).

Table 1 confirms each prediction. In Chang's design, DML-DiD is unbiased (bias below 0.001 in absolute value, MCSE 0.001) and covers at 0.960 ($K = 2$) and 0.956 ($K = 5$). Abadie's plug-in estimator is biased upwards by 0.071 (MCSE 0.002), about 1.7 of its own standard deviations. This reproduces the contrast Chang uses to motivate DML-DiD. The unadjusted DiD, however, is also unbiased, covers at 0.948, and is the most precise of the estimators (SD 0.017 against 0.024 for DML-DiD).[1] In this design, adjusting for covariates buys nothing.

Once the covariates enter the trend, the unadjusted DiD is biased by 1.559–1.566 across N, matching the predicted 1.563, and never covers. DML-DiD with post-Lasso first steps removes this bias only gradually. At $N = 200$ the outcome Lasso selects on average 0.21 of the five trend coefficients, so $\hat{\ell}_{1k}$ is close to a constant, and the adjustment comes almost entirely from the propensity score, where the Lasso selects 1.3 covariates. The bias is 0.565 (MCSE 0.028), a third of the unadjusted bias, coverage is 0.604, and the distribution of $\tilde{\theta}$ has heavy tails (SD 0.64 against a robust SD of 0.52; SE/SD 0.84). At $N = 1000$ and 2000, on average 3.07 and 3.95 of the trend coefficients are selected, the bias falls to 0.096 and 0.033, and coverage rises to 0.884 and 0.946 (MCSE 0.014 and 0.010). Without the refit, as in Chang's specification, the selected coefficients are shrunk towards zero, and DML-DiD removes much less of the confounding: the bias is 1.431, 0.776 and 0.490, and coverage is at most 0.004. In every cell, B_1 tracks the bias (with the refit 0.617, 0.094 and 0.036 against 0.565, 0.096 and 0.033; without it 1.429, 0.781 and 0.486 against 1.431, 0.776 and 0.490), as Lemma 4 and Lemma 25 require. The oracle covers at 0.946–0.964 throughout, so the shortfall comes from the first steps, not from the variance estimator. Abadie's plug-in estimator remains biased by 0.95–1.43.

Where: Replace "Digest" (p. 22).

What D1 shows. First, the bias of DML-DiD is the product term B_1 of Lemma 4, in every cell and for both versions of the first steps. Second, Chang's simulation design contains no confounding: the unadjusted DiD is unbiased there and more precise than DML-DiD. His simulation shows that a plug-in first step harms Abadie's estimator and that the orthogonal score repairs this. It does not show that DML-DiD removes confounding. Third, once the covariates do confound the trend, DML-DiD needs a large sample even in an exactly sparse

design with five relevant covariates: coverage is 0.60 at $N = 200$ and 0.88 at $N = 1000$, and reaches the nominal level at $N = 2000$. Fourth, the refit matters. Without it, the shrinkage of the selected coefficients leaves half of the confounding at $N = 1000$ and a third at $N = 2000$, and coverage is zero.

5.3 D2 and D3 (pp. 22–23)

What to fix in the text that is there.

- D3 is never defined. In your code D2 is the design with aligned signs and D3 the one with random signs.
- "$X \sim N(0, 1)$" should be $X \sim N(0, I_p)$; "$g_0 = \Lambda(X'\delta)$" should be $\Lambda(X'\gamma)$ **(wrong as printed)**; "let both the outcome and the propensity nuisance follow a trend" is not what is meant.
- "$R = 200$": the CSV files have $R = 500$.
- "the estimates with refitted penalties": see D1.
- Priors: for $(2, 2)$, "coverage is expected to approach 1" should be "approach the nominal level". The priors contain no numbers, although the theory gives them.
- "Results" and "Digest" are empty.

Where: Replace the whole of "Second and third design" (pp. 22–23).

Second and third design: decaying coefficients

D2 and D3 address the case in which the nuisance functions are not sparse in the dictionary the researcher uses: every covariate matters a little, as in the dense models of Giannone et al. (2021). The coefficients decay polynomially, and the rate of decay moves the design from dense to approximately sparse. Designs with polynomially decaying coefficients are standard in this literature (Belloni et al. 2014; Belloni et al. 2017; Athey et al. 2018); the combination of different decay rates for the two nuisances follows Smucler et al. (2019, Appendix D).

Design. $X_i \sim N(0, I_p)$ with $p = N/2$, $g_0(X) = \Lambda(X'\gamma)$ and $\ell_{10}(X) = 1 + X'\delta$, where $|\gamma_j| \propto j^{-\alpha_g}$, $|\delta_j| \propto j^{-\alpha_\ell}$ and $\|\gamma\|_2 = \|\delta\|_2 = 1$, so that the strength of selection and of the trend is the same in every cell and only the decay differs. The outcome is $Y_i(1) - Y_i(0) = \ell_{10}(X_i) + D_i\theta_0 + \varepsilon_i$ with $\theta_0 = 1$ and $\varepsilon_i \sim N(0, 1)$. In D2 the signs are aligned, $\gamma_j, \delta_j > 0$. In D3 the signs are independent random draws, fixed once per cell and nested across N, so that the leading coefficients are the same for every N. The decay rates are $(\alpha_g, \alpha_\ell) \in \{(0.7, 0.7), (2, 2), (0.8, 1.5)\}$, $N \in \{500, 1000, 2000, 4000\}$, and $K = 5$.

Priors. With coefficients that decay like $j^{-\alpha}$, the Lasso error is of order $N^{-(1-e)/2}$ with $e = 1/(2\alpha)$, up to logarithms (Smucler et al. 2019, Example 10). Chang's condition, an error below $N^{-1/4}$ for each first step, holds if $\alpha > 1$ for both. The product condition of Proposition 6 holds if $1/\alpha_g + 1/\alpha_\ell < 2$. The three cells are chosen accordingly: $(0.7, 0.7)$ violates both conditions, $(2, 2)$ satisfies both, and $(0.8, 1.5)$ satisfies the product condition but not Chang's. Four predictions follow.

1. By Lemma 4, the bias equals B_1 in every cell, and by Proposition 9, B_1 is given by the errors of the two coefficient vectors.
2. The bias relative to the standard error, $\sqrt{N}\,|\text{bias}|$, behaves like $N^{(e_g+e_\ell-1)/2}$. It grows in $(0.7, 0.7)$, with exponent 0.21, so coverage goes to zero (Proposition 7); it falls in $(2, 2)$,

with exponent -0.25, so coverage approaches the nominal level (Proposition 8); and in $(0.8, 1.5)$ the exponent is -0.02: the product condition holds, but so narrowly that the bias should stay roughly constant relative to the standard error at any realistic N.

3. By (2), the bias is the sum of $\gamma_j\delta_j$ over the covariates that both fits omit. With aligned signs every term is positive; with random signs the terms partly cancel. The bias should therefore be smaller in D3 than in D2 wherever many covariates are omitted.

4. Without the refit, the shrinkage of the selected coefficients adds to the bias in every cell.

As a benchmark, the bias of the unadjusted DiD is $0.826\,\gamma'\delta$ in all cells, by the Remark of Section 4.1 with $p_0 = 0.5$ and $\mathbb{E}[\Lambda'(X'\gamma)] = 0.207$.

				$N = 500$		$N = 1000$		$N = 2000$		$N = 4000$		Slope	
(α_g, α_ℓ)	Signs	$\gamma'\delta$	Unadj.	Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.	Est.	Pred.
A. DML-DiD, post-Lasso													
(0.7, 0.7)	aligned	1.00	0.83	0.511	0.04	0.445	0.00	0.377	0.00	0.320	0.00	0.27	0.21
(0.7, 0.7)	random	−0.52	−0.43	−0.146	0.78	−0.091	0.80	−0.079	0.72	−0.058	0.73	–	–
(2, 2)	aligned	1.00	0.83	0.065	0.93	0.045	0.90	0.018	0.95	0.016	0.93	−0.25	−0.25
(2, 2)	random	−0.85	−0.71	0.073	0.91	0.036	0.91	0.014	0.94	0.010	0.96	–	–
(0.8, 1.5)	aligned	0.87	0.72	0.180	0.61	0.121	0.66	0.083	0.66	0.057	0.66	−0.05	−0.02
(0.8, 1.5)	random	0.76	0.63	0.095	0.84	0.022	0.93	0.001	0.95	0.000	0.95	–	–
B. DML-DiD, Lasso (no refit)													
(0.7, 0.7)	aligned	1.00	0.83	0.729	0.00	0.648	0.00	0.566	0.00	0.488	0.00	0.31	0.21
(0.7, 0.7)	random	−0.52	−0.43	−0.354	0.17	−0.243	0.18	−0.173	0.15	−0.128	0.10	–	–
(2, 2)	aligned	1.00	0.83	0.279	0.23	0.181	0.29	0.121	0.31	0.079	0.41	−0.11	−0.25
(2, 2)	random	−0.85	−0.71	−0.133	0.72	−0.044	0.87	−0.009	0.91	0.003	0.94	–	–
(0.8, 1.5)	aligned	0.87	0.72	0.402	0.02	0.301	0.00	0.220	0.00	0.161	0.00	0.06	−0.02
(0.8, 1.5)	random	0.76	0.63	0.316	0.15	0.204	0.17	0.123	0.28	0.067	0.49	–	–

Table 3: D2 (aligned signs) and D3 (random signs): bias and coverage of the nominal 95% interval of DML-DiD, $R = 500$, $\theta_0 = 1$. $\gamma'\delta$ and the bias of the unadjusted DiD are given for $N = 4000$. "Slope" is the slope of $\log(\sqrt{N}|\text{bias}|)$ on $\log N$ and its predicted value $(e_g + e_\ell - 1)/2$. The largest MCSE of a bias in the table is 0.006.

			Post-Lasso				Lasso (no refit)			
(α_g, α_ℓ)	Signs	N	Bias	B_1	Formula	Both omit	Bias	B_1	Inner	Both omit
(0.7, 0.7)	aligned	500	0.511	0.528	0.520	0.655	0.729	0.734	0.900	0.630
(0.7, 0.7)	aligned	4000	0.320	0.320	0.317	0.357	0.488	0.486	0.543	0.336
(0.7, 0.7)	random	500	−0.146	−0.138	−0.139	−0.176	−0.354	−0.354	−0.418	−0.171
(0.7, 0.7)	random	4000	−0.058	−0.057	−0.057	−0.063	−0.128	−0.124	−0.147	−0.060
(2, 2)	aligned	500	0.065	0.074	0.070	0.076	0.279	0.280	0.307	0.069
(2, 2)	aligned	4000	0.016	0.014	0.014	0.015	0.079	0.077	0.076	0.014
(2, 2)	random	500	0.073	0.062	0.060	0.067	−0.133	−0.137	−0.152	0.060
(2, 2)	random	4000	0.010	0.008	0.008	0.009	0.003	0.004	0.003	0.009
(0.8, 1.5)	aligned	500	0.180	0.197	0.190	0.240	0.402	0.408	0.498	0.230
(0.8, 1.5)	aligned	4000	0.057	0.055	0.055	0.064	0.161	0.158	0.177	0.060
(0.8, 1.5)	random	500	0.095	0.088	0.091	0.123	0.316	0.311	0.377	0.116
(0.8, 1.5)	random	4000	0.000	0.001	0.002	0.001	0.067	0.069	0.078	0.001

Table 4: D2 and D3: bias of DML-DiD, B_1 on the test draw, and the approximations of Section 4.1. "Formula" is Proposition 9 evaluated at the fitted coefficients, "Inner" is (1) and "Both omit" is (2).

Results. Tables 3 and 4 show the following.

1. *The bias is B_1.* In every cell the bias of DML-DiD and B_1 agree to within 0.02 at $N = 500$ and 0.004 at $N = 4000$, and Proposition 9 reproduces B_1 to within 0.01 for post-Lasso. The standard errors themselves are accurate: SE/SD is between 0.92 and 1.05 in all cells. The failures of coverage below are failures of centring, not of the variance estimator.

2. *Dense in both nuisances.* In $(0.7, 0.7)$ with aligned signs, post-Lasso leaves 62% of the bias of the unadjusted DiD at $N = 500$ and 39% at $N = 4000$, where the Lasso selects on average 5 of the 2,000 covariates for the propensity score and 6 for the trend. $\sqrt{N}\,|\text{bias}|$ grows from 11 to 20 (slope 0.27, against a predicted 0.21), and coverage is 0.04 at $N = 500$ and zero thereafter, as Proposition 7 predicts.

3. *Sparse in both.* In $(2, 2)$, the bias of post-Lasso falls from 0.065 to 0.016, the slope of $\sqrt{N}\,|\text{bias}|$ is -0.25, as predicted, and coverage is between 0.90 and 0.95.

4. *Inside the product region, outside Chang's condition.* In $(0.8, 1.5)$ with aligned signs, $\sqrt{N}\,|\text{bias}|$ barely moves, from 4.0 at $N = 500$ to 3.6 at $N = 4000$ (slope -0.05, against a predicted -0.02), and coverage stays at 0.61–0.66. Proposition 8 applies to this cell, but its conclusion is not visible at these sample sizes: an exponent of -0.02 would require N to grow by many orders of magnitude before the bias becomes small relative to the standard error.

5. *Alignment.* With random signs, the contributions of the omitted covariates partly cancel. In $(0.8, 1.5)$ the bias of post-Lasso is 0.022 at $N = 1000$ and vanishes from $N = 2000$, with coverage 0.93–0.95, against 0.66 with aligned signs. In $(0.7, 0.7)$ the share of the unadjusted bias that remains at $N = 4000$ falls from 39% to 13%, but coverage is still only 0.72–0.80. In $(2, 2)$, where few covariates are omitted, alignment makes little difference. For random signs I do not report slopes: the bias falls below its Monte Carlo error or changes sign as more covariates are selected, so $\log|\text{bias}|$ does not measure a rate.

6. *The refit.* Without the refit, the bias is 1.4 to 7 times larger in the cells with aligned signs, and coverage is at most 0.41, even in $(2, 2)$. For post-Lasso, the sum over the covariates that both fits omit, (2), tracks B_1 and over-predicts it by 3–24%. Without the refit it misses the shrinkage of the selected coefficients (0.069 against a B_1 of 0.280 in $(2, 2)$ at $N = 500$), and only the inner product (1) tracks B_1.

What D2 and D3 show. When both nuisances are dense in the researcher's dictionary, DML-DiD with Lasso first steps removes only part of the confounding and its confidence interval does not cover, although the standard errors are accurate. The product condition describes the limit correctly, but close to its boundary the limit is irrelevant in practice: a design that satisfies the condition has a coverage of two thirds at every sample size considered. Whether the remaining bias is large depends on whether the omitted covariates push selection and trends in the same direction, which the researcher cannot check from the selected model. The refit matters in every cell.

Note: $\mathbb{E}[\Lambda'(X'\gamma)] = 0.207$ is implied by your notebook output (unadjusted bias 0.826 at $\gamma'\delta = 1$, $p_0 = 0.5$); print `e_dlam` once to confirm the third decimal. If your supervisor wants D2/D3 shorter, move Table 4 and result 6 to the appendix; the main text then needs only Table 3 with the aligned rows (`tables/d23_table_main.tex`) and results 1–5.

5.4 D4 (pp. 23–24)

What to fix in the text that is there.

- "investigates the how DML-DiD grapples in the setting of Kolesár et al." needs rewriting; "categoricals and dummies are included in the structural model, where only one of the representations of the covariate matrix is truly sparse" is not accurate: several codings are sparse, and one is not.
- The setup stops in the middle of a sentence ("Category"), numbers the categories 1–3 where your code and my tables use 0–2, and does not give C, the shares, the codings, D4b, N, R or the estimators.
- "By corollary ??" is Corollary 14; "By Lemma ??" is Lemma 10. There is a stray "I" on p. 24.
- The "Digest" is my list of priors from the chat, pasted as running text ("P10, ... predicted at 0.045, 0.119 and 0.195 for = 0.05 ... equals B and the bias"). It is not a digest and several symbols were lost.

Where: Replace the whole of "Fourth design" (pp. 23–24).

Fourth design: the same truth in different codings

D4 addresses the case in which the nuisance functions are sparse in one coding of the covariates but not in another. It transfers the categorical examples of Kolesár et al. (2026, Section 3.3) to DML-DiD. The truth is fixed and only its coding changes: each replication draws one data set and one fold split, and every coding is estimated on the same data with the same folds, so differences between codings are pure effects of the representation.

Design. A categorical covariate Z with C categories enters both nuisances, $g_0 = \Lambda(-0.25 + \gamma_Z)$ and $\ell_{10} = 1 + \delta_Z$, and $Y_i(1) - Y_i(0) = \ell_{10} + \theta_0 D_i + \varepsilon_i$ with $\theta_0 = 1$ and $\varepsilon_i \sim N(0,1)$. Every dictionary also contains $N/10$ irrelevant $N(0,1)$ covariates, $N \in \{1000, 2000, 4000\}$, $R = 200$ and $K = 5$.

- *D4a (reference category).* $C = 20$. Category 0 is atypical in both selection and trends, $(\gamma_0, \delta_0) = (1,1)$; category 1 only in selection, $(1,0)$; category 2 only in trends, $(0,1)$; categories 3–19 are typical, $(0,0)$. Category 0 has share $\pi_0 \in \{0.05, 0.15, 0.30\}$ and the other categories share the rest equally. The codings are dummies with reference category 3, 1, 2 or 0.
- *D4b (ordered categories).* $C = 40$ ordered categories with equal shares, like age bins. Selection and trends jump in the middle, $\gamma_c = \mathbf{1}\{c \geq 20\}$ and $\delta_c = 0.75 \cdot \mathbf{1}\{c \geq 20\}$. The codings are thresholds, $\mathbf{1}\{Z \geq c\}$ for $c = 1, \ldots, 39$, and dummies with the lowest category as reference.

Besides DML-DiD with and without the refit, I report two estimators that do not penalise the dummies: OR-DiD on the full dictionary, and DML-DiD with the dummies *forced in*, where the Lasso penalises only the irrelevant covariates.

Priors.

1. *Reference category.* By Corollary 14, only reference category 0, which is atypical in both nuisances, creates bias. If the Lasso gives categories 1 and 2 their own levels and pools category 0 with the typical ones, the corollary predicts a bias of 0.045, 0.119 and 0.195 for $\pi_0 = 0.05$, 0.15 and 0.30. This exceeds the bias of the unadjusted DiD (0.039, 0.108 and 0.180), because categories 1 and 2 contribute to the unadjusted bias with the opposite sign. References 1, 2 and 3 are atypical in at most one nuisance and should be harmless.
2. *Pooling.* By Proposition 13, the bias equals the pooling formula evaluated at the selected columns, in every cell.

3. *Ordered categories.* Threshold coding needs one selected column and should be unbiased. Dummy coding needs 20 dummies, each resting on 2.5% of the sample. As long as the Lasso cannot select them, it pools across the jump, and the bias approaches that of the unadjusted DiD, 0.184.

4. *Unpenalised dummies.* By Lemma 10, OR-DiD gives the same estimate under every coding and is unbiased, at the price of a larger variance, since it estimates all $N/10+C-1$ coefficients. Forcing the dummies in makes DML-DiD invariant to the reference category as well. I expect it to be unbiased, with a variance between that of DML-DiD with a harmless coding and that of OR-DiD.

5. *The range grows with N.* Where a coding is biased, the bias stays while the standard error shrinks, so the studentised range across codings, $\mathcal{R}(\mathcal{A})$, should grow with N.

6. *The refit.* Without the refit, the shrinkage of the selected coefficients adds bias, also for the harmless codings.

	Post-Lasso							No refit		Selected	
	Bias	MCSE	SD	SE/SD	Cov.	B_1	Pool	Bias	Cov.	$\hat{g}$	$\hat{\ell}$
$\pi_0 = 0.05$*: unadjusted bias 0.039, Corollary for reference 0: 0.045*											
Reference 3 (typical)	−0.002	0.002	0.031	1.07	0.97	−0.001	0.000	0.025	0.91	1.9	1.9
Reference 1 (selection only)	−0.002	0.002	0.030	1.08	0.96	−0.001	0.000	0.024	0.92	0.9	1.9
Reference 2 (trend only)	−0.001	0.002	0.031	1.08	0.97	−0.001	0.000	0.024	0.93	1.9	0.9
Reference 0 (both)	0.044	0.002	0.031	1.06	0.77	0.046	0.045	0.041	0.76	0.9	1.0
Dummies forced in	−0.002	0.002	0.031	1.07	0.97	0.000	–	−0.002	0.97	all	all
OR-DiD	−0.001	0.002	0.035	–	–	–	–	–	–	–	–
Unadjusted DiD	0.039	0.002	0.031	1.07	0.81	–	–	–	–	–	–
$\pi_0 = 0.15$*: unadjusted bias 0.108, Corollary for reference 0: 0.119*											
Reference 3 (typical)	−0.001	0.002	0.032	1.06	0.98	0.000	0.000	0.023	0.90	1.8	2.0
Reference 1 (selection only)	−0.001	0.002	0.031	1.07	0.97	0.000	0.000	0.023	0.90	1.0	2.0
Reference 2 (trend only)	−0.001	0.002	0.032	1.05	0.97	0.000	0.000	0.022	0.91	1.8	1.0
Reference 0 (both)	0.117	0.002	0.033	1.04	0.06	0.115	0.118	0.110	0.09	0.7	1.0
Dummies forced in	−0.002	0.002	0.032	1.05	0.99	0.000	–	−0.002	0.99	all	all
OR-DiD	0.000	0.003	0.037	–	–	–	–	–	–	–	–
Unadjusted DiD	0.107	0.002	0.032	1.05	0.10	–	–	–	–	–	–
$\pi_0 = 0.30$*: unadjusted bias 0.180, Corollary for reference 0: 0.195*											
Reference 3 (typical)	0.000	0.002	0.031	1.08	0.97	0.000	0.000	0.022	0.93	1.7	2.0
Reference 1 (selection only)	0.000	0.002	0.030	1.11	0.98	0.000	0.000	0.022	0.93	1.0	2.0
Reference 2 (trend only)	0.000	0.002	0.032	1.09	0.97	0.000	−0.001	0.021	0.92	1.7	1.0
Reference 0 (both)	0.189	0.002	0.033	1.04	0.00	0.190	0.190	0.181	0.00	0.4	0.9
Dummies forced in	0.000	0.002	0.032	1.09	0.97	0.000	–	0.000	0.96	all	all
OR-DiD	0.001	0.003	0.038	–	–	–	–	–	–	–	–
Unadjusted DiD	0.179	0.002	0.033	1.04	0.00	–	–	–	–	–	–

Table 5: D4a at $N = 4000$, $R = 200$, $\theta_0 = 1$. The label of each coding says in which nuisance its reference category is atypical. "Pool" is the formula of Proposition 13 evaluated at the selected columns; "No refit" is the same Lasso fit plugged in without refitting; "Selected" is the average number of dummies the Lasso selects for the propensity score and the trend. OR-DiD is identical under all four reference categories.

		Reference 3		Reference 0		Forced in		
π_0	N	Bias	Cov.	Bias	Cov.	Bias	Cov.	Range
0.05	1000	0.024	0.94	0.038	0.91	−0.002	0.95	0.12
	2000	0.007	0.94	0.043	0.83	0.000	0.94	0.86
	4000	−0.002	0.97	0.044	0.77	−0.002	0.97	1.47
0.15	1000	−0.015	0.93	0.106	0.66	−0.005	0.94	1.75
	2000	−0.001	0.94	0.115	0.38	−0.002	0.94	2.51
	4000	−0.001	0.98	0.117	0.06	−0.002	0.99	3.63
0.30	1000	−0.007	0.94	0.175	0.27	−0.003	0.94	2.68
	2000	−0.003	0.96	0.182	0.04	−0.003	0.96	3.87
	4000	0.000	0.97	0.189	0.00	0.000	0.97	5.64

Table 6: D4a across N: bias and coverage of DML-DiD (post-Lasso) with a typical reference category, with the reference category that is atypical in both nuisances, and with the dummies forced in. "Range" is the median over replications of the studentised range across the four reference categories, in standard errors of the estimate with reference 3.

Results. Tables 5–7 confirm the priors.

1. *Only the reference category that is atypical in both nuisances creates bias.* At $N = 4000$ its bias is 0.044, 0.117 and 0.189 for $\pi_0 = 0.05$, 0.15 and 0.30 (MCSE 0.002), against 0.045, 0.119 and 0.195 predicted by Corollary 14, and coverage is 0.77, 0.06 and 0.00. With references 1, 2 and 3 the bias is at most 0.002 in absolute value and coverage is 0.96–0.98. A reference category that is atypical in only one nuisance is harmless: this is the categorical form of the product structure of Lemma 4.

2. *The pooling formula gives the bias.* B_1 agrees with the bias to within 0.010 in every cell, and to within 0.003 in D4a at $N = 4000$. The formula of Proposition 13, evaluated at the selected columns, is a population quantity that ignores the sampling error of the refit; it agrees with the bias to within 0.015, with the largest gaps at $N = 1000$.

3. *The bias does not shrink with N.* For $\pi_0 = 0.30$ the bias of reference 0 is 0.175, 0.182 and 0.189 at $N = 1000$, 2000 and 4000, while its standard error falls, so coverage drops from 0.27 to 0.00 (Table 6). At $N = 4000$ the Lasso selects on average 0.4 and 0.9 of the 19 dummies. With C fixed, the dummies of the typical categories would eventually be selected, but not at these sample sizes.

4. *A bad reference category is worse than no adjustment.* With reference 0 the bias exceeds that of the unadjusted DiD (0.189 against 0.179 for $\pi_0 = 0.30$, and likewise for the other shares), because the Lasso does fit categories 1 and 2, whose contributions to the unadjusted bias have the opposite sign.

5. *When the Lasso selects almost nothing, the coding does not matter.* At $N = 1000$ and $\pi_0 = 0.05$, every coding pools nearly all categories, all codings are biased (0.024 with reference 3 and 0.038 with reference 0), and the median studentised range is 0.12. A small range is therefore not evidence that a specification is unbiased.

6. *Ordered categories.* In D4b, threshold coding is unbiased (bias at most 0.007 in absolute value, coverage 0.94–0.95). Dummy coding stays at the bias of the unadjusted DiD (0.175, 0.190 and 0.173, against 0.184), with coverage 0.28, 0.03 and 0.00.

7. *Unpenalised dummies.* OR-DiD gives the same estimate under every coding, to machine precision, and is unbiased, with a standard deviation 13–23% larger than that of DML-DiD with a harmless coding at $N = 4000$. With the dummies forced in, DML-DiD

	Post-Lasso							No refit		Selected	
	Bias	MCSE	SD	SE/SD	Cov.	B_1	Pool	Bias	Cov.	$\hat{g}$	$\hat{\ell}$
$N = 1000$: median studentised range 2.58											
Thresholds	−0.005	0.005	0.070	1.00	0.94	0.001	0.000	0.056	0.85	1.7	1.7
Dummies, reference 0	0.175	0.005	0.069	0.99	0.28	0.183	0.183	0.176	0.28	0.0	0.0
Dummies forced in	−0.014	0.009	0.122	0.88	0.96	−0.009	–	−0.008	0.93	all	all
OR-DiD	−0.002	0.006	0.079	–	–	–	–	–	–	–	–
Unadjusted DiD	0.175	0.005	0.069	0.98	0.26	–	–	–	–	–	–
$N = 2000$: median studentised range 3.70											
Thresholds	0.007	0.003	0.047	1.04	0.95	0.001	0.000	0.042	0.87	2.0	1.9
Dummies, reference 0	0.190	0.003	0.045	1.10	0.03	0.180	0.183	0.192	0.01	0.1	0.0
Dummies forced in	0.006	0.004	0.051	1.11	0.98	−0.001	–	0.006	0.97	all	all
OR-DiD	0.006	0.004	0.053	–	–	–	–	–	–	–	–
Unadjusted DiD	0.192	0.003	0.043	1.11	0.01	–	–	–	–	–	–
$N = 4000$: median studentised range 5.06											
Thresholds	0.000	0.002	0.033	1.03	0.95	0.000	0.000	0.019	0.93	1.9	1.9
Dummies, reference 0	0.173	0.002	0.032	1.07	0.00	0.176	0.181	0.180	0.00	0.6	0.1
Dummies forced in	0.000	0.002	0.035	1.04	0.95	0.000	–	0.000	0.94	all	all
OR-DiD	−0.001	0.003	0.038	–	–	–	–	–	–	–	–
Unadjusted DiD	0.181	0.002	0.032	1.07	0.00	–	–	–	–	–	–

Table 7: D4b: ordered categories with a jump in the middle, $R = 200$, $\theta_0 = 1$. The bias of the unadjusted DiD is 0.184 in the population. Columns as in Table 5.

is unbiased in D4a (bias at most 0.005 in absolute value, coverage 0.94–0.99), and its standard deviation is within 6% of that of DML-DiD with a harmless coding. Forcing in is not free when the categories are small: in D4b at $N = 1000$, where each of the 40 bins holds 25 observations, the standard deviation is 0.122 against 0.070 for threshold coding, and the standard error understates it (SE/SD 0.88). At $N = 2000$ and 4000 the difference is 9% and 6%.

8. *The range grows with N.* The median studentised range across the four reference categories grows from 0.12, 1.75 and 2.68 at $N = 1000$ to 1.47, 3.63 and 5.64 at $N = 4000$ for the three shares, and from 2.58 to 5.06 across the two codings of D4b. This is the order of magnitude that Kolesár et al. (2026) document in their applications, where such choices move sparsity-based estimates by two standard errors or more.

9. *The refit.* Without the refit, the harmless codings are biased by 0.02–0.07, with coverage 0.82–0.93.

What D4 shows. The coding of a categorical covariate matters for DML-DiD exactly where Proposition 13 says it should: when the Lasso pools categories that differ in both selection and trends. The damage grows with the share of the pooled category, it does not fade over the range of N considered, and it can exceed the bias of making no adjustment at all. A researcher can avoid the choice by not penalising the dummies, which costs little unless the categories are very small, or, for ordered categories, by threshold coding. D4 uses a single categorical covariate with a known structure. Section 6 asks how much these choices matter in Chang's application.

Note: (i) The forced-in files contain only `forced_ref3` (and `forced_ref0` for D4b). The invariance of the forced-in estimate to the reference category is therefore a consequence of Lemma 10, not something the tables show. If you want to show it, run one cell with a second reference; since your code implements "unpenalised" as a penalty divided by 100, the two estimates will agree to about four decimals, not exactly, and the text should say so. (ii) The three tables take most of two pages. If space is short, keep Tables 6 and 7 in the main text and move Table 5 to the appendix.

6 Inserts: Section 6, Chang's application

6.1 State of the section

Section 6 of v13 (pp. 24–25) is still the plan from v11: an opening paragraph in which five citations are broken, the summary of Sequeira's paper, a paragraph of intentions ("I intend to conflate the covariate matrix . . . "), a note in Danish, and an empty heading "Sparsity in DML".

Update, 14 November. You have since run Chang's nine covariates with your own first steps (50 splits, $K = 5$): -3.157 (SE 3.481) with post-Lasso and -12.682 (SE 2.263) without the refit. At your suggestion I then ran Chang's original replication code (`Table 2/DMLML.R` in his repository). It reproduces his published estimate to every digit, and taking it apart changes what Section 6 should say. Part 6.2 is rewritten accordingly. The scripts behind the numbers are in `replication/`.

Correction. The first version of this file said that Chang's estimate, -5.222, equals the unadjusted difference in differences, -5.228, because "his first steps do essentially no adjustment". That was wrong. His Lasso fits select on average 3.7 of the nine covariates for the propensity score and 2.0 for the outcome; the two numbers agree by coincidence. The claim is removed from the inserts below and from the conclusion.

The text below is complete except for the numbers in [orange], which must come from your run. The numbers in panels A and C of Table 8 are from my runs today; rerun them yourself before you use them (Part 6.2 says how).

Small fixes in the paragraph you keep ("Is trade liberalization a briber's utopia?"): "a 8-year window" should be "an 8-year window"; and the paragraph describes the tariff and trade-flow data but not the data Chang uses, the bribe payments recorded for individual shipments. Add one sentence on where they come from and how treatment is defined (a tariff change in 2008 for the product), from Sequeira's paper.

6.2 Opening, and what the application can show

Where: Replace the opening paragraph of Section 6 (p. 24).

> Chang (2020) illustrates DML-DiD with the study of Sequeira (2016) on tariff liberalisation and corruption. In this section I use his application to ask how far the choices of representation studied in Section 4 move the DML-DiD estimate in real data. The data are repeated cross-sections, so the estimator is based on the score ψ_2.

Where: After the summary of Sequeira's paper. Replace "Chang uses only nine covariates . . . " and the Danish note.

> **Chang's specification.** The outcome is $\log(1+\text{bribe})$ for a shipment, treatment is a change in the tariff of the product in 2008, and the period indicator is an indicator for shipments after 2008. Chang's replication code builds dummies for clearing agents and product groups, but the covariate matrix uses only its first nine columns: the post-2008 indicator, which is the period indicator T itself; the 2007 tariff; the log value per ton; indicators for differentiated, agricultural and perishable goods and for large firms; the day of the week of arrival, entered as a number from 1 to 7; and an indicator for monitoring. With $N = 1084$ complete cases, this is hardly a high-dimensional setting.

Recreating Chang's estimate. Chang reports a DML-DiD estimate of −5.222 with a standard error of 2.647. His replication code reproduces both numbers exactly.[2] The code follows Definition 2 with $K = 2$ folds, penalties chosen by 10-fold cross-validation with the one-standard-error rule, and no refit, and it averages the estimate over 100 random splits. In three respects it does something that the paper does not describe. First, observations with a fitted propensity score outside $(0.03, 0.97)$ are dropped. Second, both nuisance functions are predicted without the intercept that the Lasso has fitted, $\hat{g} = \Lambda(X'\hat{\beta})$ and $\hat{\ell}_2 = X'\hat{\beta}$. The fitted propensity scores then average 0.27, while 16% of the shipments are treated. Third, within each fold, the code discards every score that is at least as large as the absolute value of the smallest score in the fold before it averages them. This removes on average 5 of about 530 scores per fold, all of them positive.

			Single splits	
	Estimate	SE	Min	Max
A. Chang's procedure: cross-validated penalty, no refit, $K = 2$, 100 splits				
Chang (2020, Table 2)	−5.222	2.647		
His replication code	−5.222	2.647	−13.71	7.62
… all scores kept	4.308	4.592	−5.35	18.34
… fitted intercepts included	−12.105	1.711	−15.53	−9.17
… intercepts included and all scores kept	−10.091	2.377	−14.64	0.47
B. This thesis: plug-in penalty, clipping, $K = 5$, 50 splits				
Lasso (no refit)	−12.682	2.263	−13.69	−11.28
Post-Lasso	−3.157	3.481	−6.15	3.14
C. Without first steps				
Difference in differences of the four group means	−5.228			
Score ψ_2, first steps depend on the period only	−13.33			
Score ψ_2, constant first steps	−18.67			

Table 8: DML-DiD in Chang's application with his nine covariates, $N = 1084$. Panel A: mean over the 100 splits and standard error as computed by Chang's code, with one step of the code changed at a time. Panel B: median over the splits and the standard error of Chernozhukov et al. (2018, Section 3.4).

Panel A of Table 8 shows what these steps do. The third one decides the sign: with all scores kept, the same code gives +4.3. With the fitted intercepts included, it gives −12.1, and −10.1 when all scores are kept as well. The published estimate thus depends on two steps of the code that the paper does not describe. Implemented as the paper describes it, with Chang's own tuning, DML-DiD gives −10.1 with a standard error of 2.4, about twice the published estimate.

Panel B shows my implementation with the same nine covariates. Without the refit it gives −12.7, close to panel A with the intercepts included, and close to what the score gives when the first steps depend on the period only (−13.3, panel C): the shrunken Lasso fits adjust for little more than the period. With the refit, the estimate is −3.2 with a standard error of 3.5, and single splits range from −6.1 to 3.1.

What the application can show. For one data set and one set of nine covariates, the estimates in Table 8 range from +4 to −13. Two features of the data add to this. First, the estimator for repeated cross-sections assumes that both periods are samples from the same population (RCS(i)). In the data, the treated share is 0.40 among the 140 shipments before 2008 and 0.13 among the 944 after, and five of the eight clearing agents appear only after 2008. This is why the estimate depends so strongly on how the period indicator, which Chang

includes among the covariates, is fitted: with constant first steps the score gives -18.7, and with first steps that depend on the period only it gives -13.3, because the score then weights the post-period difference between treated and controls by $\Pr(T = 1 \mid D = 1)/\Pr(T = 1) = 0.78$ and the pre-period difference by $\Pr(T = 0 \mid D = 1)/\Pr(T = 0) = 2.46$. Second, 82.3% of the outcomes are zero, and an average effect on $\log(1 + Y)$ depends on the units in which Y is measured (Chen and Roth 2024). I therefore do not interpret the level of any of these estimates. I keep Chang's specification and use the application to measure how far the coding of the covariates moves the estimate within it.

Note: (i) Panels A and C are my runs. To make the numbers yours: for the second row, run Chang's `DMLML.R` unchanged with the line `RNGkind(sample.kind = "Rounding")` added before `DMLML(Y,D,p,T)`; for the other rows of panel A, use `replication/chang_variants.R` (exact, in R) or the Python port `replication/chang_port.py`, which follows the same steps with other random splits and gives numbers within about one point of these. (ii) The statement about the paper is a statement about code, and it is easy to check: the lines are `beta1hat=fit$beta`, `ellhat2=p[index1,]%*%beta2hat` and `s=s[which(s<abs(min(s)))]` in `DMLML.R`; neither the paper nor its online appendix mentions trimming or the removal of scores. Still, show the table to your supervisor before you build the section on it, and keep the wording descriptive, as above. (iii) "for large firms": the variable is `dfs`; check its definition in Sequeira's paper before you describe it. (iv) -5.228 is close to Chang's -5.222 by coincidence; do not comment on it.

6.3 Codings and results

Where: After the previous insert.

Codings. I add the product group of the shipment to Chang's nine covariates, as dummies. The product groups aggregate the HS 4-digit codes of the shipments. At the 4-digit level, 107 of the 131 codes in the sample contain only treated or only untreated shipments, so 4-digit dummies would violate overlap, and I leave the clearing agents out for the reason given above. One product group contains 3 shipments, all treated. Without controls, its untreated trend is not identified, and by the caveat to Lemma 10 even the unpenalised estimate would depend on the coding, so I drop it ($N = 1081$, 13 groups). The 2007 tariff almost determines treatment: products with tariffs of 0 or 2.5 are never treated, and products with tariffs of 3, 4 or 10 always are. Dummies or polynomials in the tariff would push the estimated propensity score to 0 or 1, and I leave it linear. Agricultural products are almost never treated (2 of 522 shipments), so the comparison effectively rests on non-agricultural shipments.

I compare three sets of specifications: Chang's nine covariates; the same with product-group dummies, once for each of the 13 choices of reference group; and the same with the dummies forced in, which by Lemma 10 does not depend on the reference group. All estimates use the same **[100]** random sample splits with $K = 5$, and I report the median over splits with the standard error of Chernozhukov et al. (2018, Section 3.4), which accounts for the variation across splits.

Results. Table 9 reports the estimates. The starting point is the post-Lasso estimate with Chang's nine covariates, -3.2 (SE 3.5, Table 8). With product-group dummies, the propensity Lasso selects the dummies of **[three]** groups, **[vegetables, machinery and transportation]**, whose treated shares (0.67, 0.67 and 0.05) differ most from the others. **[Nine]** of the other ten groups, as reference, yield the same pools and therefore the same estimate ([+5.2]). Only the three groups whose dummies are selected change the estimate substantially when they serve as the reference. The default choice in applied work, the largest group, is transportation, which is atypical in both selection and trends (Table 2); it gives the estimate furthest from the benchmark with the dummies forced in ([+8.1] against [−2.0]), as Corollary 14 leads one to

expect. Across reference groups, the estimate moves by [12] points, or [1.3] standard errors of the specification with transportation as the reference.

The standard errors are large, because the 53 treated shipments from before 2008 dominate the score, and the level of the estimate is not credible given the change in composition between the periods. What the exercise shows is that, in Chang's application, the choice of reference group alone moves DML-DiD by more than a standard error, that the sign of the estimate depends on it, and that the references that matter are the ones Proposition 13 points to.

Note: The orange numbers are from my own run with 20 splits and an earlier version of the first steps. Replace all of them, and check each sentence against your table: in particular which dummies are selected, and whether the forced-in estimate is still negative. If it is not, delete "that the sign of the estimate depends on it". Table 9: one row per specification (Chang's nine covariates; dummies forced in; each reference group), columns: post-Lasso estimate (SE), Lasso estimate (SE), number of group dummies selected in $\hat{g}$ and in $\hat{\ell}_2$; below it the range over reference groups in standard errors.

6.4 Centring

The introduction (p. 2) and the list in Section 4 name three choices of representation: the reference category, the coding of categorical variables, and where a variable is centred before it is squared or interacted. The thesis has theory and simulations for the first two and nothing for the third. Either add the centring check to Section 6 or remove the third choice from the introduction and mark it in Section 4 as a choice you do not study. If you run it, the text is:

Centring. The third choice of Section 4 is where a variable is centred before it is squared or interacted. I add squares and pairwise interactions of the continuous covariates to Chang's specification, which gives a dictionary of [39] columns, and centre the covariates at zero, at their means or at their medians before forming them. The three dictionaries span the same space. The estimate moves by [0.24] standard errors across the three centrings, against [0.06] when only a single square is added. Kolesár et al. (2026) find ranges above one standard error from centring, in dictionaries with several hundred columns. In a dictionary of the size that Chang's sample allows, centring is a second-order concern compared with the coding of the product groups.

Note: Linear terms are exactly invariant to centring in your implementation (standardisation and an unpenalised intercept), so centring can only matter through the squares and interactions. Say this in one sentence if you include the paragraph.

7 Insert: conclusion

There is no conclusion in v13, although the introduction announces one. The draft below answers the two questions of the introduction in order and uses only results that are in the thesis. The paragraph on the application depends on Section 6.

Where: New Section 7.

7 Conclusion

This thesis asked how DML-DiD with Lasso first steps behaves when the sparsity it relies on fails, and which choices of representation make it fail.

The answer to the first question follows from one identity. The bias of DML-DiD is the product of the errors in the propensity score and in the untreated outcome trend (Lemma 4).

Three consequences follow. First, the conditions for DML-DiD can be stated for the product: one nuisance may be far from sparse if the other is sparse enough, which is weaker than Chang's requirement on each first step, and when the product condition fails, coverage can go to zero (Propositions 7 and 8). Second, when the Lasso selects nothing, DML-DiD is the unadjusted DiD, and in general the bias comes from the covariates that drive both selection and trends and that both first steps omit (Proposition 9). Third, because the outcome nuisance is a trend, it does not matter whether the levels of the outcome are sparse.

The answer to the second question is that the choices that matter are those that make the Lasso pool units that differ in *both* selection and trends. With categorical covariates, a post-Lasso fit is constant on groups of categories, and the bias is the covariance of selection and trends within these groups (Proposition 13). A reference category that is atypical in only one of the two is harmless; one that is atypical in both creates a bias in proportion to its share of the sample (Corollary 14). With many small categories, a single atypical reference category therefore costs little. What matters is the size of the category, and the common default of using the largest category as the reference is the risky one.

The simulations confirm these results and add three observations. The product condition describes the limit, but close to its boundary the limit is irrelevant: a design that satisfies it had a coverage of two thirds at every sample size up to 4,000. A bad reference category can leave more bias than no adjustment at all, and the bias does not fade as the sample grows, so the estimates under different codings move apart by up to 5.6 standard errors. And the refit after selection matters throughout: Chang's specification, which plugs in the penalised fits, had a coverage near zero in designs where post-Lasso was close to the nominal level. The simulations also show that the design Chang uses to motivate DML-DiD contains no confounding, so that the unadjusted DiD is unbiased there.

In Chang's application, **[the choice of reference group for the product-group dummies moves the estimate by 1.3 standard errors, and the default choice, the largest group, gives the estimate furthest from the invariant benchmark.]** The level of the estimate cannot be interpreted. The composition of the sample changes between the periods, and for Chang's own nine covariates the estimate ranges from $+4$ to -13 depending on details of the implementation: his published estimate of -5.2 rests on two steps of his code that the paper does not describe, and without them the same procedure gives -10.1.

For applied work, the results suggest four practices. Report DML-DiD under alternative codings of the controls, together with their range; a large range indicates that sparsity fails in at least one of them, although a small range does not prove that it holds. Do not penalise the dummies of a categorical control when overlap allows it: the estimate is then invariant to the reference category, at a small cost in precision unless the categories are very small. Refit after selection. And compare the estimate with the unadjusted DiD: if the two coincide, the first steps have adjusted for nothing.

The analysis has limits. It covers the two-period, two-group design, and the theory is worked out for repeated outcomes, with the extension to repeated cross-sections only sketched. It concerns Lasso first steps; other learners, such as random forests, are not invariant to the representation either, but the pooling argument does not apply to them. The simulations use independent Gaussian covariates and a single categorical covariate. Finally, the thesis offers no formal test of sparsity for DML-DiD: the tests of Kolesár et al. (2026) compare with OLS, and an analogue based on the range across codings, or on the comparison with unpenalised first steps, is a natural next step, as is the extension to staggered treatment timing.

Note: "up to 5.6 standard errors" is the median range in D4a at $N = 4000$ and $\pi_0 = 0.30$. "coverage near zero in designs where post-Lasso was close to the nominal level" refers to D1 at $N = 2000$ (0.000 against 0.946). Adjust the last paragraph to what you want to claim; "random forests" is my addition and not a result of the thesis, so cut it if you prefer.

8 Literature

8.1 Claims checked against the sources

I checked the following against the texts of the papers (Chang 2020 and its appendix; Chernozhukov et al. 2018; Kolesár et al., arXiv v5 of 13 May 2026; Smucler et al. 2019; Wüthrich and Zhu 2023; Belloni et al. 2012; Belloni and Chernozhukov 2011; Belloni, Chernozhukov and Wei 2016).

Page	Claim in v13	Verdict
2, 13	Chang: valid if first steps converge faster than $N^{-1/4}$; Assumption 3.1 bounds each error by one ε_N	Correct (his Theorem 3.1 and Assumption 3.1).
7–9	Scores, Definition 1, Assumption 3, variance estimators	Agree with Chang, apart from the typos in M1–M6. His Definition 3.1 has the $\hat{p}_k$ typo described in M4.
10	Chang chooses the penalty by cross-validation and does not refit	Correct ("10-fold cross validations", p. 189).
10	Plug-in penalty "Belloni and Chernozhukov (2011)"	Chang cites Belloni et al. (2012) for the rule you use. Cite that.
10	Post-Lasso: Belloni et al. (2014, JEP)	Wrong source. Belloni and Chernozhukov (2013, *Bernoulli*).
11	Kolesár et al.: definition (7) and condition (8)	Correct (their (2) and (3)).
11	Kolesár et al., footnote 13: product condition under cross-fitting	Correct. The footnote adds that such assumptions are "likewise sensitive to normalization choices", which is your research question; quote it (M11).
4	Kolesár et al. develop two tests	Correct (a residual test and a Hausman test, their Section 5).
12	Lemma 4 is the mixed bias property of Smucler et al., eq. (3)	Correct. Their term for the class is "bilinear influence function".
12	Assumption 5: Belloni and Chernozhukov (2011, Cor. 1); van de Geer (2008, Thm 2.2, Ex. 1); Smucler et al. (Thm 9, Remark 7)	Not the estimators you use. Replace by M10.
13	"stricter than necessary (Smucler et al., 2019; Zimmert, 2020)"	Smucler et al.: correct. I could not check Zimmert. The closest source is Chernozhukov et al. (2018, Remark 5.2); cite it.
13–14	Chernozhukov et al. (2018), eq. (5.4), (3.7), (3.8), Theorem 3.2	Correct. Their Assumption 5.1 for the ATT already has the product-only rate condition, and Remark 5.2 states $s^g s^m \ll N$. See Part 9.
5, 15	Wüthrich and Zhu: under-selection when coefficients are below half the penalty; double under-selection	Correct.
1	Snow, 1849	1855 (second edition). See Part 3.
22	D2 design: Smucler et al., Appendix D	Correct for $(0.8, 1.5)$ and for the idea of different decay rates; $(0.7, 0.7)$ and $(2, 2)$ are your own cells.

Not checked. The session could not reach arXiv, so I could not check: Rodrigues and Hatfield (2026) and the description "graphical framework … equivalence between parallel trends and equi-confounding"; the NBER number of Goldsmith-Pinkham (2026); Hünermund et al. (2023); Zimmert (2020), described as "builds on the same setup" (the paper is independent of Chang's and contemporaneous, so "considers the same problem" is safer); and whether Callaway et al. (2025), Caetano and Callaway (2026) and Karim and Webb (2026) have been published in a journal in the meantime. Check these yourself.

8.2 References to add

- Belloni et al. (2012): the penalty loadings (Section 5) and the rate of the outcome Lasso (Assumption 5).
- Belloni and Chernozhukov (2013): post-Lasso.
- Belloni et al. (2016): logit Lasso, its penalty and its rate.
- Belloni et al. (2014): post-double-selection, the estimator Kolesár et al. study; also polynomially decaying designs.
- Farrell (2015): the product condition for doubly robust estimators.
- Belloni et al. (2017) and Athey et al. (2018): polynomially decaying designs (D2/D3).
- Hirano et al. (2003): estimated versus true propensity score (D1, footnote).
- Sur and Candès (2019): bias of the unpenalised logit when p/n is not small (caveat to Lemma 10).
- Chen and Roth (2024): $\log(1+Y)$ (Section 6).
- Snow (1855) and Angrist and Pischke (2009).

The entries are in `review/v13_refs.bib`.

8.3 The reference list as printed (pp. 25–27)

- "Angrist, J. D. (n.d.)": Angrist and Pischke (2009), Princeton University Press.
- "Van der Vaart, A. (n.d.). *Asymptotic statistics* (Vol. 1998)": year 1998, and the DOI is printed twice in the link.
- "Belloni, A., & Chernozhukov, V. (2011, September 2)": it is a chapter in *Inverse Problems and High-Dimensional Estimation* (Springer, 2011); cite the chapter, not the arXiv date.
- Smucler et al.: the title prints "`$\ell_1$`" as raw LaTeX.
- Sequeira: "southern africa" needs capitals (protect them with braces in the `.bib`).
- "Van De Geer" should be "van de Geer".
- "(Eilers et al.)" in Definitions 21–22 has no year and no entry; Pitman (1993) is cited only for Markov and Chebyshev. Both disappear if you cut that material (Part 9).
- Belloni et al. (2014, JEP) is no longer cited once the post-Lasso reference is corrected.

9 Stating the obvious? What is new and what is not

You asked whether the thesis only states the obvious. It does not, but it distributes its emphasis the wrong way round: the known results are presented as contributions and proved at length, and the original ones are not mentioned in the introduction.

Result	Status	What to do
Lemma 4 (mixed bias)	Known (Smucler et al.; implicit in Chernozhukov et al.).	Keep, as now, with the attribution. Add the proof (M9).
Proposition 6 (product of sparsity indices)	Follows in one line from Assumption 5. Known: Chernozhukov et al. (2018, Remark 5.2), Farrell (2015), Kolesár et al., footnote 13.	Keep, but say that it is known (M11). Delete the two sentences after it ("Both parts follow directly ... goes to zero exactly when $s \log p/N$ does"), which restate it.
Proposition 8 (rate double robustness)	Theorem 5.1 of Chernozhukov et al. applied to the change in outcomes.	Present it as that (M13). Its point is that Chang's condition is stronger than needed.
Proposition 7 (rate failures)	(a) is routine. (b)–(c), an explicit sequence for which coverage goes to zero, I have not seen written down; Chernozhukov et al. assert sharpness without proof.	Yours. Keep. Say that (b) is a worst case and link it to D2/D3 (M12).
Remark (no adjustment)	Elementary.	Keep as a remark. Its value is the benchmark numbers it gives for the simulations (1.563, $0.826\,\gamma'\delta$, 0.184).
Proposition 9 (Gaussian bias)	Yours. The "both omit" form is the DML-DiD counterpart of Wüthrich and Zhu.	Keep and emphasise. Use the simpler constants (M15).
Lemma 10 (invariance)	Elementary; Kolesár et al. note it for OLS.	Keep, short. The caveats (separation, categories without controls) are the part that is not obvious and that Section 6 needs.
Lemma 11, Proposition 12	Algebra. The decomposition is never used again: no table reports its three terms and the bound is never evaluated.	See below.
Proposition 13, Corollary 14 (pooling, reference category)	Yours, and the most original part of the thesis.	Put them in the introduction (Part 3).
D1	Partly a replication. New: Chang's design has no confounding; why Abadie's estimator is biased there; the role of the refit.	Say these three things explicitly (Part 5).
D2–D4	Yours.	Part 5.

Proposition 12. Two options. (i) Reduce it: state the identity in one display in the text, keep the observation that treated observations are unaffected by the propensity fit (so that a change of coding acts through the weights of the controls and through the outcome fit), keep the definition of $\mathcal{R}(\mathcal{A})$, move Lemma 11 to the appendix, where the proof of Lemma 27 uses it, and drop the bound. (ii) Use it: report the three terms for one pair of codings in D4 or in Section 6. I recommend (i); it saves a page in the text and a page and a half in the appendix.

The appendix. The first four pages (pp. 28–32) are textbook material, and most of it is not used in any proof.

- Cut Definition 15 (limits), the empty Definition 16, and Definitions 20–22 (epsilon functions, o and O for functions). Nothing refers to them.
- Cut the proof of Chebyshev's inequality from Markov's (Definitions 17–18, Lemma 19). You use Chebyshev once, for $O_k = O_P(n^{-1/2})$. Write "by Chebyshev's inequality" there.
- Keep the o_P/O_P notation (Definition 23) as a short paragraph, and Definition 24 as a

lemma with its source (Chernozhukov et al. 2018, Lemma 6.1). Markov, Chebyshev and Definition 24 are results, not definitions.

- "Proof of zero mean adjustment term" is two lines. Put it in Section 3.2 as one sentence: $\mathbb{E}[c_j \mid X] = 0$ because $\mathbb{E}[D \mid X] = g_0$.
- Cut the "Walk-through of lasso estimation": it derives the logistic log-likelihood, which is already displayed in Section 3.3.
- Cut the paragraph that apologises to the reader (p. 28).

This removes about four pages and loses nothing an examiner would look for.

In the text. Sentences that tell the reader nothing can go: the second half of the paragraph after Proposition 6; "This is the upper bound of the error in estimation, and the realized error can be much smaller" (p. 38); "g_N is thus a legitimate probability" (p. 38) can be one clause.

10 Coherence

1. **The two questions of the introduction are never answered in one place.** The conclusion in Part 7 does that.
2. **The two kinds of failure.** The introduction distinguishes failure (1), sparse in some basis but not in the researcher's, and failure (2), not sparse in any basis. Section 4.1 is about (2) and Section 4.2 about (1); D2/D3 test (2) and D4 tests (1). Say so in the first sentence of each of these parts. The inserts for Section 5 do it for the designs.
3. **Section 4 opens with two other "questions"** (p. 10: "The first question to emerge is whether sparsity applies to DML-DiD … The second question to emergu … "). They are not the research questions, and the first sentence is ungrammatical. Suggested replacement:

> In this section, I bring the critique of Kolesár et al. (2026) to the estimator of the previous section. DML-DiD itself does not assume sparsity: it needs first-step estimators that converge fast enough. Sparsity enters through the choice of a sparsity-based first step, such as the Lasso, and DML-DiD is attractive precisely because it appears to leave the choice of covariates to the first step. But a Lasso depends on how the covariates are written down, and Kolesár et al. (2026) document the following choices, none of which affects OLS. I state all results for repeated outcomes; they carry over to repeated cross-sections with ψ_2 and ℓ_{20}, as noted after Lemma 4 and at the end of the proof of Proposition 7.

4. **Repeated outcomes and repeated cross-sections.** The theory and all simulations are for repeated outcomes; the application uses repeated cross-sections. The last sentence of the insert above and the first paragraph of Section 6 make the link.
5. **Lasso and post-Lasso.** Section 3.3 says you refit, Assumption 5 speaks of "lasso first steps", Section 5 reports both. Write "Lasso or post-Lasso first steps" in Assumption 5, and call the two versions the same thing everywhere: "post-Lasso" and "Lasso (no refit)".
6. $p \asymp N$**.** Section 4 assumes $p \asymp N$. D1–D3 have $p = N/2$. D4 has a fixed number of categories, so asymptotically the Lasso would select every relevant dummy; D4 is about what happens before that. Result 3 of D4 says so; keep that sentence.
7. **Table 1 does not match the list of estimators**, and D3 is undefined. Fixed by the inserts.
8. **Headings.** "Priors" has a Bayesian meaning; "Predictions" is the usual word. "Digest" could be "What D1 shows". I kept "Priors" in the inserts to match your headings; change it if you agree.

9. **The Remark has no number**, which is why "as per ." is empty.

10. **There is no abstract** in v13. Check what your study programme requires (abstract, front page, word count).

Notation used for more than one thing.

Symbol	Meanings in v13	Suggestion
c, c_j	adjustment terms c_1, c_2 in the scores (p. 7); constants c_1, c_2 in Proposition 9; terms c_1, c_2, c_3 in Proposition 12; rate exponent c in Proposition 7; penalty constant $c = 1.1$; constant fit $\hat{\ell} \equiv c$; category c; coverage c	$\bar{c}_1, \bar{c}_2$ in Proposition 9; (I)–(III) in Proposition 12; exponents b_g, b_ℓ; coverage in words
δ	trend coefficients; $\delta_g = g' - g$ (Lemma 11); δ_g, δ_ℓ errors (Lemma 27); a constant in the proof of Proposition 7(c)	Δg in Lemma 11; another letter in the proof
γ	propensity coefficients; penalty loadings γ_{kj} (p. 10); Chang's level $\gamma \to 0$ (p. 19)	$\hat{\psi}_{kj}$ for the loadings
α	level of the penalty; level of the test; decay rates α_g, α_ℓ	acceptable; say "decay rates" each time
n	fold size; sample size in the Lasso (p. 19)	M in the Lasso
s	dimension of θ_0 (p. 7); sparsity index	drop the first
w	argument of Λ; $w(X)$ and its mean (Prop. 9); pool weight (Cor. 14); w_g, w_0 (Lemma 27); $w_i^{(A)}$ (p. 42)	$\bar{w}$ for the mean; ω in Corollary 14
$\tilde{\theta}$, $\hat{\theta}$	both used for the final estimator	$\tilde{\theta}$ (M5)
λ	$P(T = 1)$; penalty level	acceptable, already flagged on p. 10

11 Form

Unresolved references. "Lemma ??" on p. 20 and in the caption of Table 1 (Lemma 4); p. 36 and p. 42 (Lemma 11); "corollary ??" and "Lemma ??" on pp. 23–24 (Corollary 14, Lemma 10); "as per ." on p. 20 (the Remark); "6 below" on p. 12; "(9)" and the red "repeated outcomes" on p. 7; "27(ii)" on p. 13. In the appendix, "Lemma (4)" and "inequality (17)", "(19)" print like equation numbers.

Broken citations. Listed in Part 3.

Left-over notes. "INDSÆT HER" (p. 17); the stray "I" (p. 24); the Danish note and the heading "Sparsity in DML" (p. 25); the empty Definition 16.

Typos I noticed (run a spell-check as well): p. 1 "treated an untreated"; p. 2 "ML method innocuous tools", "choices that does not"; p. 3 "one nuisances", "chang uses"; p. 4 "commmon", "simultanously", "number of covariates exceed"; p. 5 "two-nusaince", "both panel and repeated outcomes" (repeated outcomes and repeated cross-sections); p. 7 "$j = \{1, 2\}$" ($j \in \{1, 2\}$); p. 8 "nuissance"; p. 9 "of of"; p. 10 "of he nuisance", "I my thesis", "the size of the set where"; p. 11 "Since DML-DiD that allows", "a sparsity-based estimators", "emergu"; p. 13 "suffiecient"; p. 14 "uandjusted", "sparse in large"; p. 15 "that escape both fits omit"; p. 16 "Anecdotically", "$\hat{g}_K$"; p. 18 "were selection" (where), "If it atypical", "intutive", "leave the coefficients", "neglectible", "standard erro", "treshhold"; p. 19 "lasso ust", "samling", "Undjusted"; p. 20 "be follow", "maintaining letting"; p. 22 "requieres", "how DML-DiD response"; p. 23 "attatched", "suffiecient", "the how"; p. 24 "selct"; p. 25 "a 8-year", "commmon", "used create", "conflate the covariate matrix" (you mean "expand"), and "Since squares and interactions of dummy variables are degenerate, I will include them" (you mean "exclude"); appendix: "supress", "Walk-trough", "estimation of of", "asusmption", "$q_p X_i$", "th whole", "$\hat{\lambda}_K$". Also write "Lasso" with a capital throughout (it

is an acronym), and choose between British and American spelling ("categorise"/"categorize", "centred"/"centered" both occur).

12 Questions for you

1. **Section 6.** The baseline and the recreation of Chang's estimate are done (Table 8). Do you want the full comparison across the 13 reference groups and the forced-in dummies (about 15×50 cross-fits), or two or three reference groups only? And are you comfortable reporting panel A of Table 8? Ask your supervisor.
2. **Centring.** Include the check in Section 6, or drop centring as a choice you study?
3. **D2/D3.** Main text with both tables, or one table in the text and the rest in the appendix? Is there a page limit?
4. **Titles.** Has your supervisor agreed to the informal title and section titles?
5. **Forced-in dummies.** Do you want to show the invariance to the reference category numerically (one extra run), or rely on Lemma 10?
6. **Deadline.** If you tell me the date, I can order the remaining work by what an examiner will weigh most.

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